MIL-M-28787/265A(EC)
2 June 1982
SUPERSEDING
MIL-M-28787/265(NAVY)
11 December 1975

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY, READ ONLY, DIGITAL

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic read only memory module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5962

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3.1.1 Application requirements. This specification establishes the general requirements and test procedures for a read only memory (ROM) module. When a Standard Electronic Modules Program (SEMP) user determines a requirement for a ROM module, the user shall prepare a design approval request (DAR) and submit the DAR to the SEMP Design Review Activity (SEMP-DRA) in accordance with MIL-STD-1378. The module user shall prepare a separate module specification. The module specification shall specify, as a minimum, the module key code as assigned by the SEMP-DRA, the firmware pattern (truth table), detailed module marking, and this specification's requirements.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.727 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I, the user's detailed specification, and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389, the user's detailed specification, and as specified herein.

3.5.1 Key code. The module key code shall be as specified in the user's detailed specification.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and in the user's detailed specification, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc, except that the limits for V_{OUT} (low-loaded) apply only for a power supply voltage of 7.0 Vdc +0, -1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

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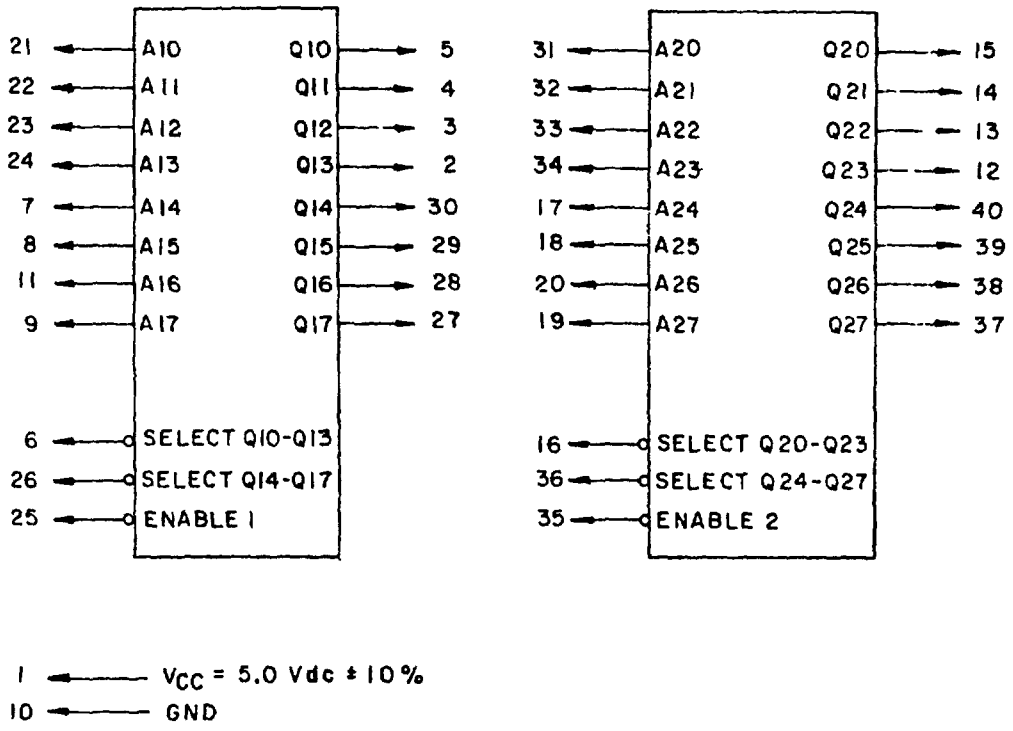


FIGURE 1. Module function diagram, read only memory.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	∞	1.0	∞	10	∞	1.0	∞	MΩ
B1-B16	V_{OUT} (low-loaded)	III	S	0.0	0.8	0.0	0.8	0.0	0.8	0.0	0.8	V
C1-C18	Input breakdown current	IV	S	0	2	0	2	0	2	0	2	mA
C19-C22	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	mA
D1-D16	Output short circuit current	V	X	-15	-90	-15	-90	-15	-90	-15	-90	mA
E1-E16	Output leakage current, low level	VI	X	-100	100	-100	100	-100	100	-100	100	µA
F1-F16	Output leakage current, high level	VII	X	-100	100	-100	100	-100	100	-100	100	µA
G1-G22	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
H1-H16	Static output voltage, low level	IX	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
J1-J16	Static output voltage, high level	X	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
* K1-K18	Input current, low level	XI	X	0	-0.56	0	-0.56	0	-0.56	0	-0.56	mA
* K19-K22	Input current, low level	XI	X	0	-0.28	0	-0.28	0	-0.28	0	-0.28	mA
L1-L18	Input current, high level	XI	X	-20	80	-20	80	-20	80	-20	80	µA
L19-L22	Input current, high level	XI	X	-20	40	-20	40	-20	40	-20	40	µA
M	Power supply current	XII	X	10	450	10	450	10	450	10	450	mA
N1-N16	Transition time (t_{TLH}), C_L = 50 pF	XIII	S	5	30	5	30	5	35	5	35	ns
N17-N32	Transition time (t_{TLH}), C_L = 150 pF	XIII	Q	5	40	5	40	5	45	5	45	ns
* N33-N48	Transition time (t_{THL}), C_L = 150 pF	XIII	S	3	30	3	30	3	35	3	35	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N49-N64	Transition time (t_{THL}), C_L = 150 pF	XIII	Q	5	40	5	40	5	45	5	45	ns
P1-P32	Enable/select recovery time (t_{ERL})	XIV	S	5	40	5	40	5	40	5	40	ns
P33-P64	Enable/select recovery time (t_{ERH})	XIV	S	5	40	5	40	5	40	5	40	ns
Q1, Q2	Galloping read access pattern	XV	X	← See table XV, no failure →								
Q3, Q4	Galloping read access pattern	XV	Q	← See table XV, no failure →								
R1-R6	Input crosstalk, low level	XVI	Q	0	400	0	400	0	400	0	400	mV
S1-S6	Input crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	mV
T	Functional test pattern	XVII	X	← See table XVII, no failure →								
U	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W18	Input capacitance	XX	Q	2	30	2	30	2	30	2	30	pF
W19-W22	Input capacitance	XX	Q	2	20	2	20	2	20	2	20	pF
W23-W38	Output capacitance	XX	Q	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787, and in the user's detailed specification.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, or D are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through W shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

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* 6.3 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	2.5 W (outputs not loaded)
Supply voltage	5 Vdc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2.0 V minimum, V_{CC} maximum
Input current, address, enable	
Low level (V_{IL} = 0.0 V)	-0.56 mA maximum
(V_{IL} = 0.4 V)	-0.50 mA maximum
High level (V_{IH} = 2.4 V)	80 µA maximum
(V_{IH} = 5.5 V)	2 mA maximum
Input current, select (program pin)	
Low level (V_{IL} = 0.0 V)	-0.28 mA maximum
(V_{IL} = 0.4 V)	-0.25 mA maximum
High level (V_{IH} = 2.4 V)	40 µA maximum
(V_{IH} = 4.5 V)	1 µA maximum
Input clamp diode voltage	
I_{clamp} = -5 mA	-0.4 V minimum, -1.5 V maximum
Output voltage	
Low level (I_{OL} = 10 mA)	0.45 V maximum
High level (I_{OH} = -900 µA)	2.4 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.45 V)	10 mA
High level (V_{OH} = 2.4 V)	-900 µA
Output short circuit current	
V_{OL} = 0.0 V, V_{CC} = 5.5 V	-15 mA minimum, -90 mA maximum
Output leakage current (high-impedance state)	
0.45 ≤ Vo ≤ 2.4 V	±100 µA maximum
Propagation delay time (see figure 7)	
t_{AA} - Address to data output	
C_L = 50 pF	80 ns maximum
C_L = 150 pF	90 ns maximum
t_{EA} - Enable/select	
C_L = 50 pF	45 ns maximum
C_L = 150 pF	55 ns maximum
Enable/select recovery time	
t_{ERL} - Low-state to high-impedance state (C_L < 10 pF)	40 ns maximum
t_{ERH} - High-state to high-impedance state (C_L < 10 pF)	40 ns maximum

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Capacitance	
Address input	30 pF maximum
Enable input	30 pF maximum
Select input	20 pF maximum
Data output	15 pF maximum

6.4 Module function. The module is organized as two independent 256-word by 8-bit read only memories. These memories may be interconnected through connector wiring to form a memory of either 256 by 16 or 512 by 8.

6.4.1 Input to output relationship. The module contains two independent 256-word by 8-bit read only memories as shown on figure 1. When the enable and select of one of the 256-word by 8-bit memories goes to the low level, the data programmed at the selected address input appears at the eight data outputs of that memory. When the enable or the select of one of the memories is high, the eight output lines of that memory are disabled to a high-impedance state. Module timing and truth table are defined on figure 7.

6.4.2 Paralleling inputs and outputs. Corresponding address inputs of the two address groups must be paralleled for nonindependent operation of the two memory sections. The 256 by 16 organization is achieved by paralleling the corresponding addresses of the two sections and connecting both circuit enable pins together as shown on figure 8. The 512 by 8 organization is achieved by paralleling the corresponding addresses of the two sections, the corresponding outputs Q10 through Q17 with Q20 through Q27, and placing an inverter between selects as shown on figure 9. A number of memory outputs can be connected to a common bus. All of the memories except one are placed in the high-impedance state, and the selected memory is enabled. When outputs of several modules are bussed together, all address lines should be addressed using synchronous clocking.

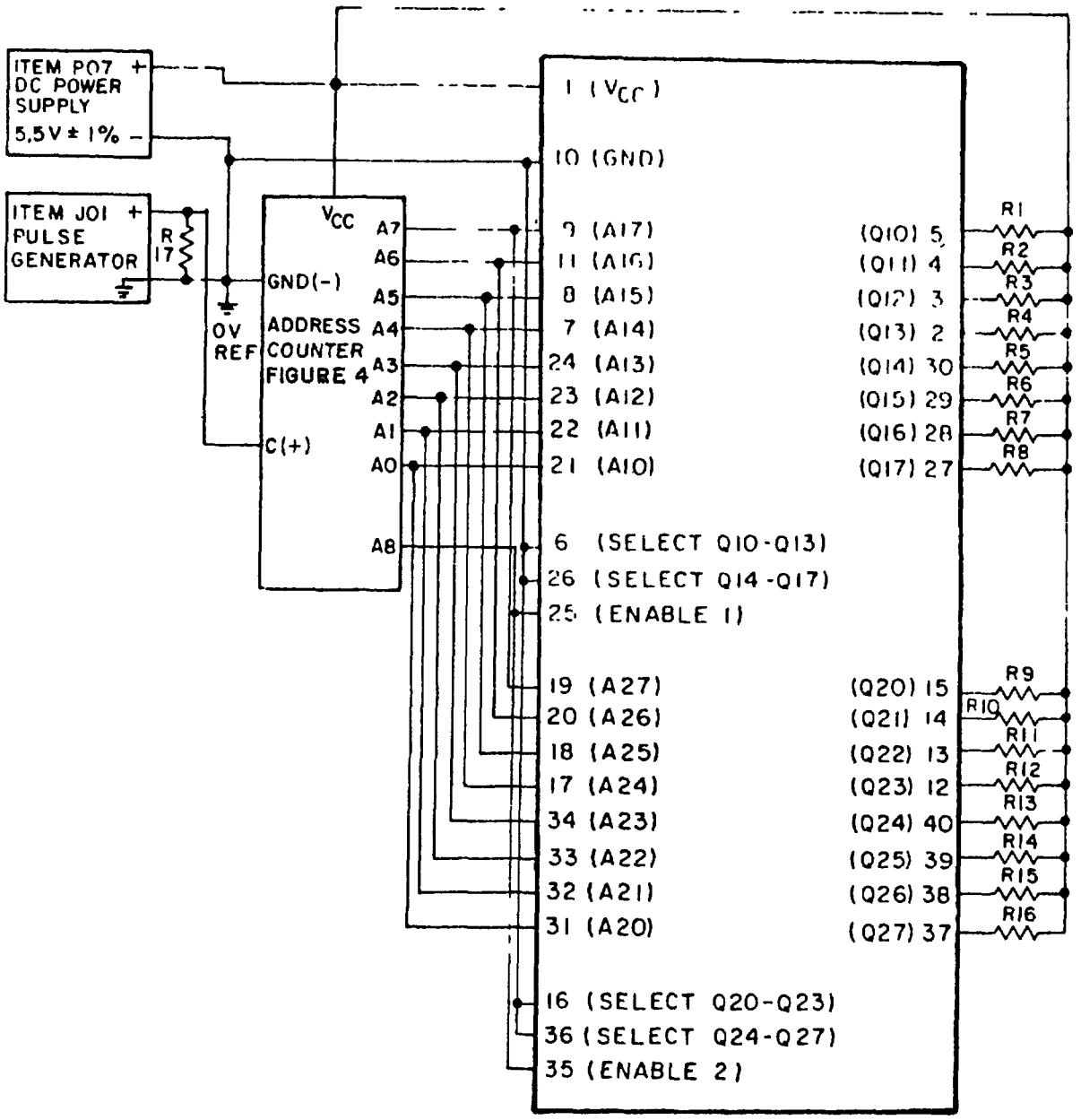
6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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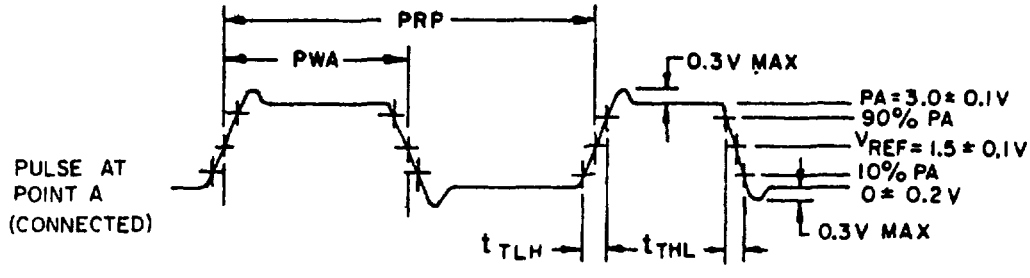
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NOTES:
1. R1 - R16 = 499Ω ±1%, 1/8 W.
2. R17 = 51Ω ±5%, 1 W.
3. See figure 3 for test circuit waveform.

FIGURE 2. Operating temperature and life test circuit.

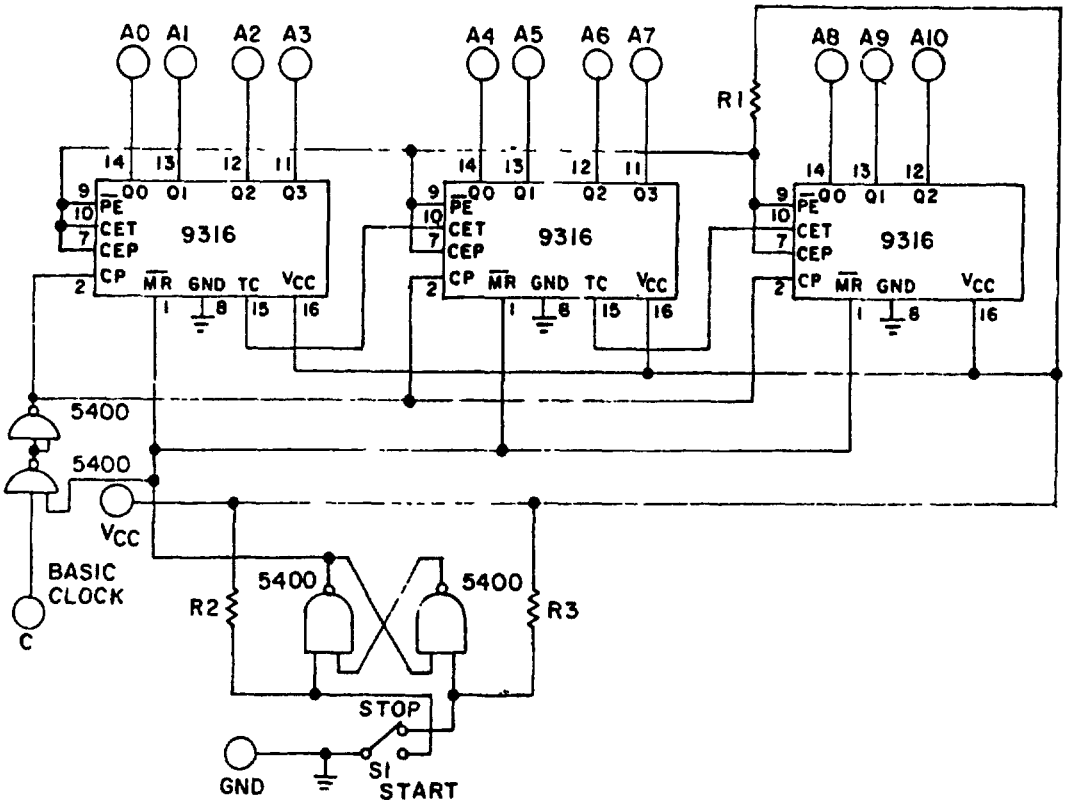
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NOTE: Pulse parameters for item J01 pulse generator:
$t_{TLH} = t_{THL} = 20 \pm 10$ ns.
PWA = 100 ±10 ns.
PRP = 200 ±10 ns.

FIGURE 3. Operating temperature , component temperature, and life test circuit waveform.

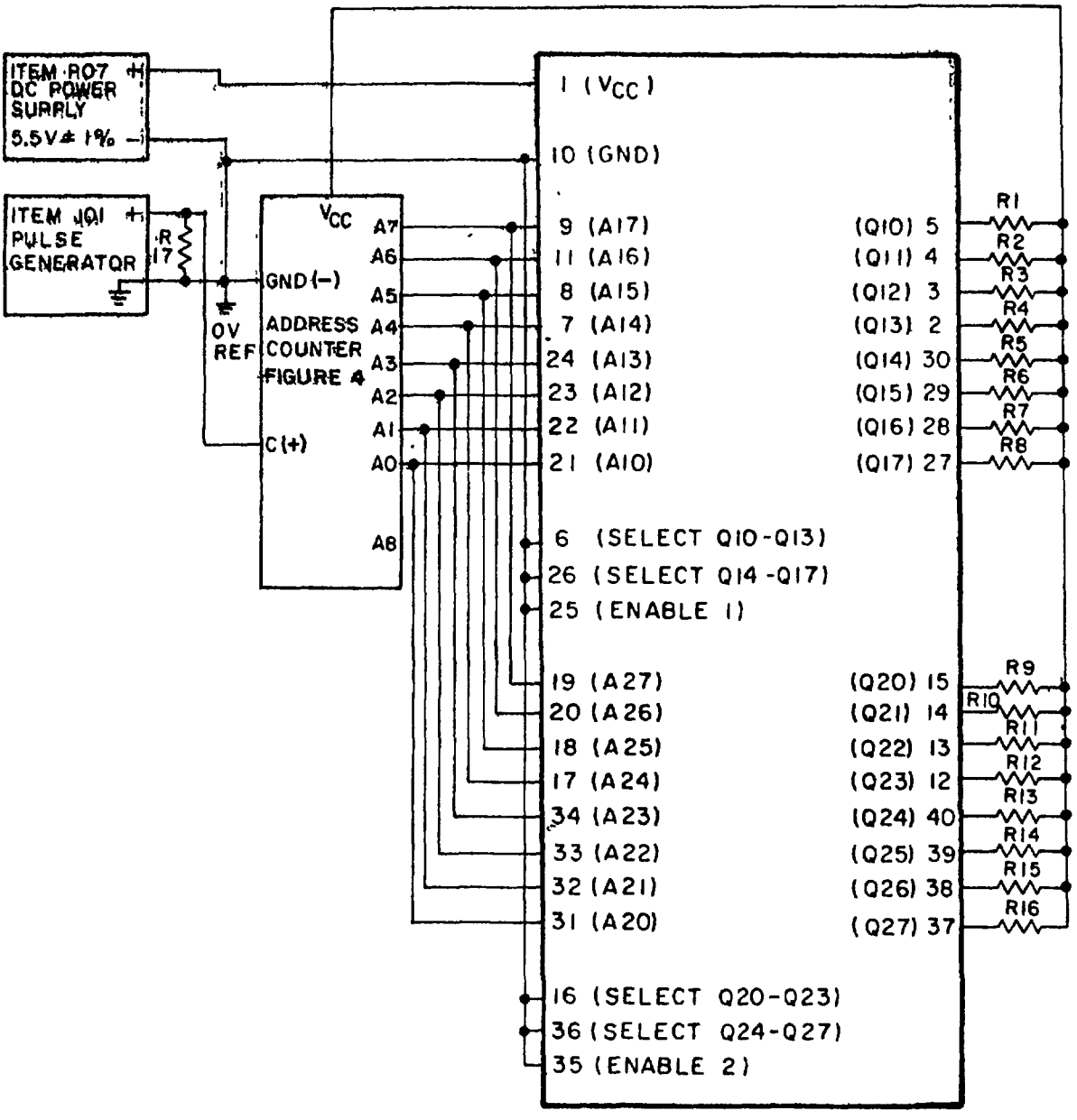
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NOTES:
1. R1-R3 = 2.2 kΩ ±1%, 1/8 W.
2. S1 = SPDT switch.
3. 3 each 9316 high-speed synchronous 4-bit binary counters.
4. 1 each 5400 quad 2-input NAND gate.

FIGURE 4. Address counter circuit.

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NOTES:
1. R1 - R16 = 499Ω ±1%, 1/8 W.
2. R17 = 51Ω ±5%.
3 Adjust item J01 waveform as specified on figure 3.

FIGURE 5. Component temperature test circuit.

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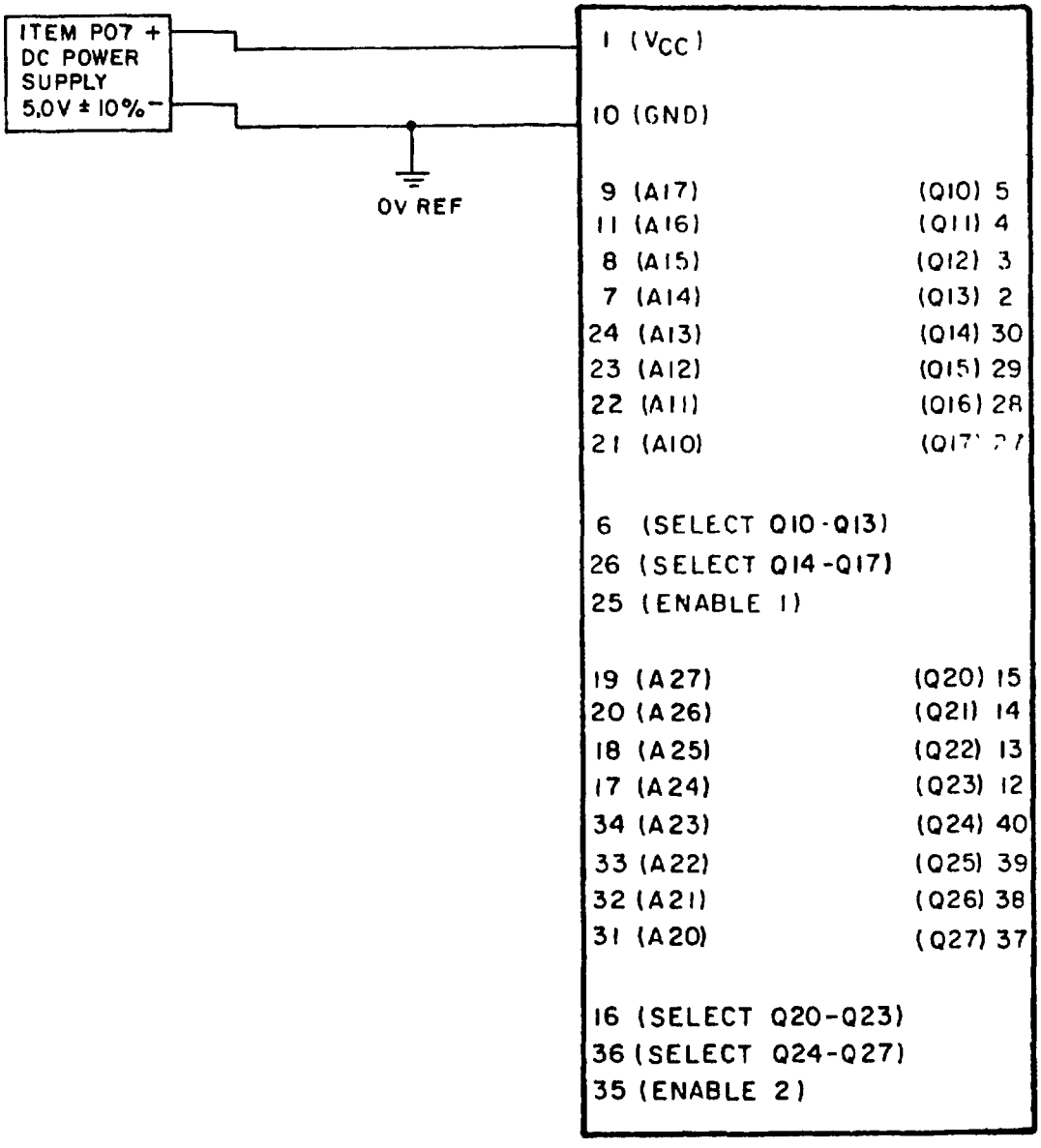
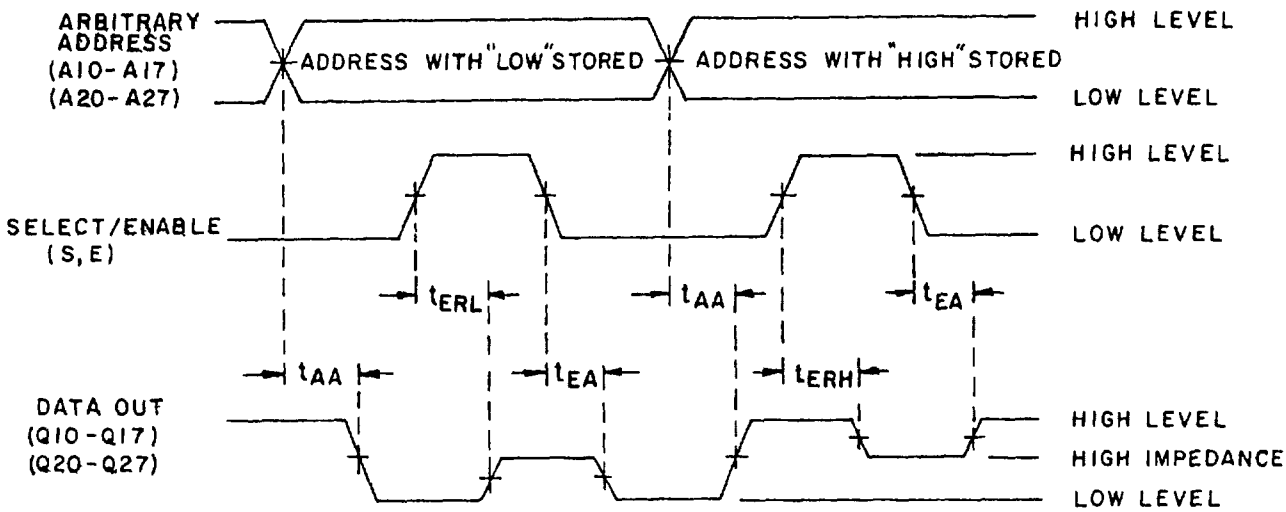


FIGURE 6. Vibration test circuit.

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DYNAMIC CHARACTERISTICS WAVEFORMS

* Module truth table.

Select(S) (Q14-Q17) (Q24-Q27)	Select(S) (Q10-Q13) (Q24-Q27)	Enable(E) (1) (2)	Outputs (Q10-Q13) (Q20-Q23)	Outputs (Q14-Q17) (Q24-Q27)
X	X	H	High Z	High Z
H	H	L	High Z	High Z
L	H	L	High Z	Read
H	L	L	Read	High Z
L	L	L	Read	Read

X = Don't care.
H = High-level input.
L = Low-level input.
High Z = High-impedance state.

FIGURE 7. Dynamic characteristics waveforms and module truth table.

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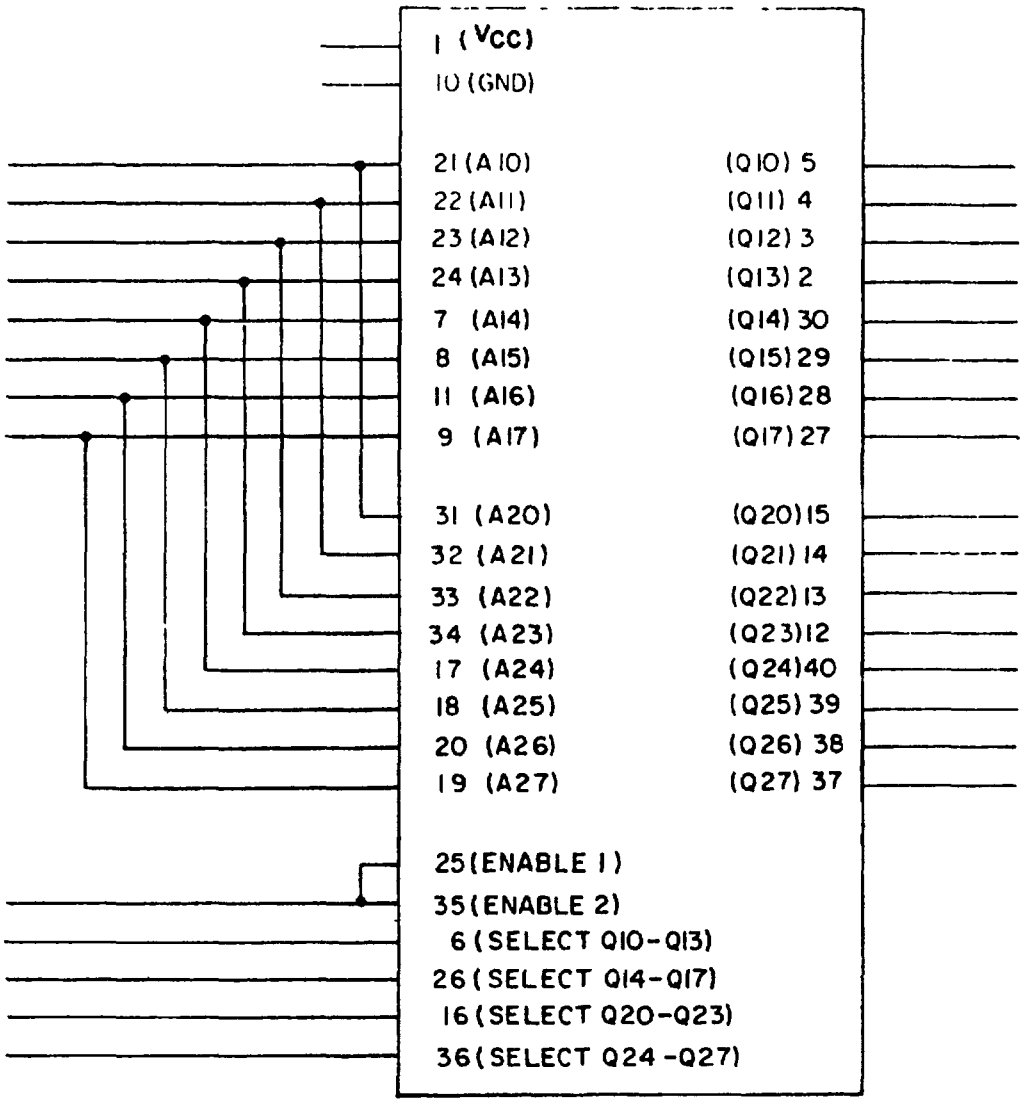


FIGURE 8. Pin interconnections for 256-word by 16-bit configuration.

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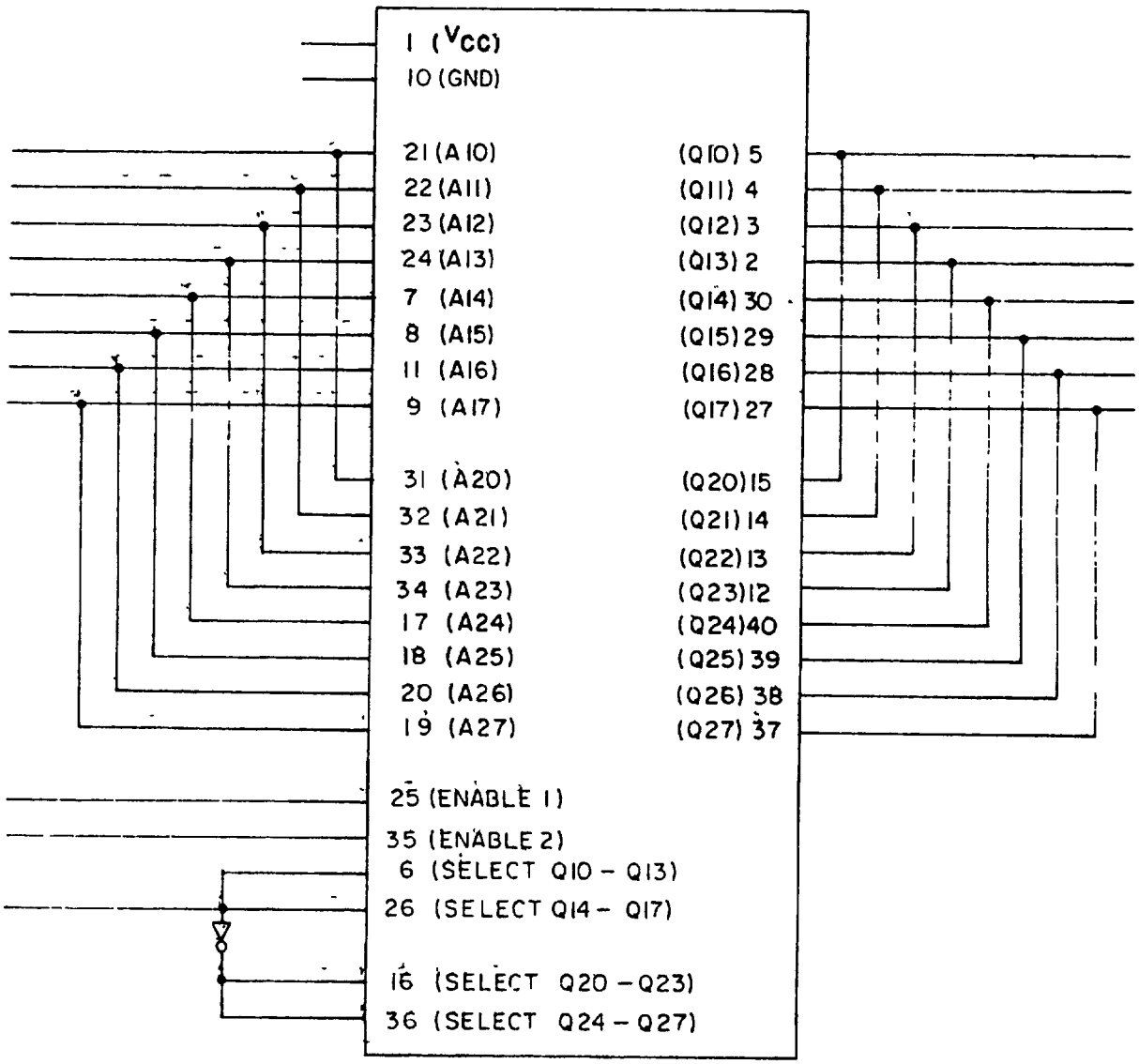


FIGURE 9. Pin interconnections for 512-word by 8-bit configuration.

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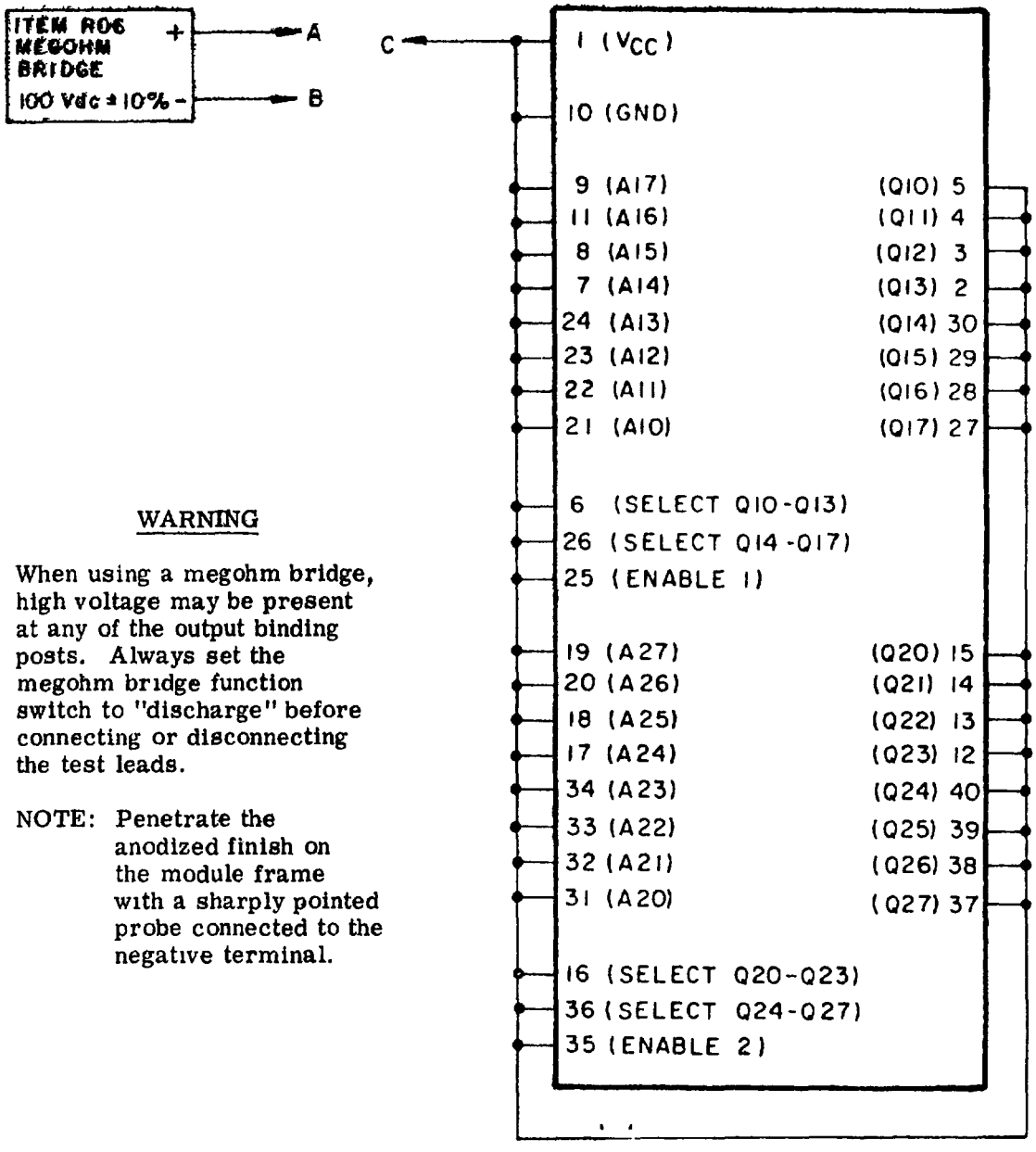
TABLE II. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	10

TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	Q10 V_{OUT} (low loaded)	5	0.0	0.8	V	11
B2	Q11	4	↓	↓	↓	↓
B3	Q12	3				
B4	Q13	2				
B5	Q14	30				
B6	Q15	29				
B7	Q16	28				
B8	Q17	27				
B9	Q20	15				
B10	Q21	14				
B11	Q22	13				
B12	Q23	12				
B13	Q24	40				
B14	Q25	39				
B15	Q26	38				
B16	Q27	37				

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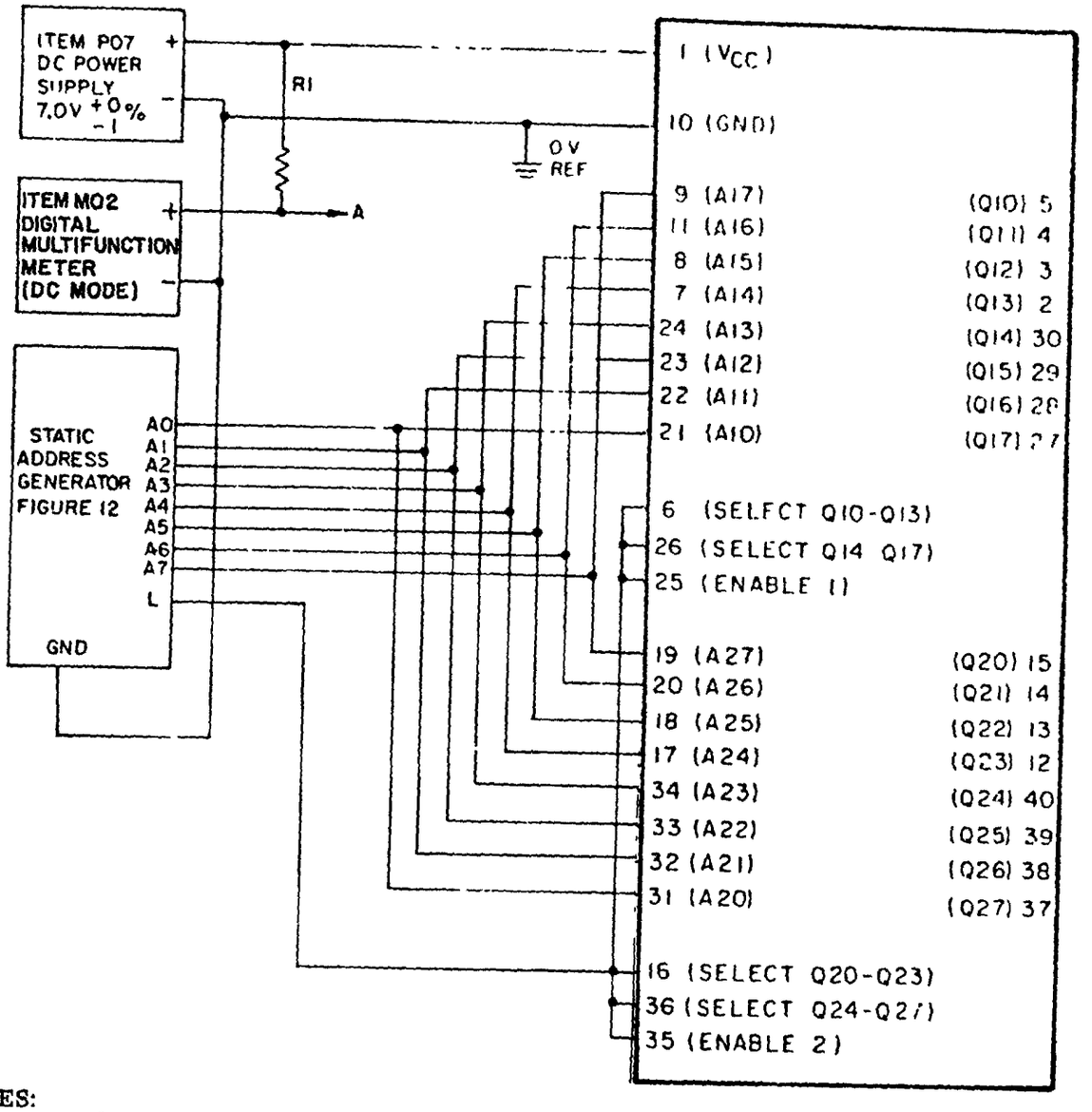
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 10. Isolation test circuit for table II.

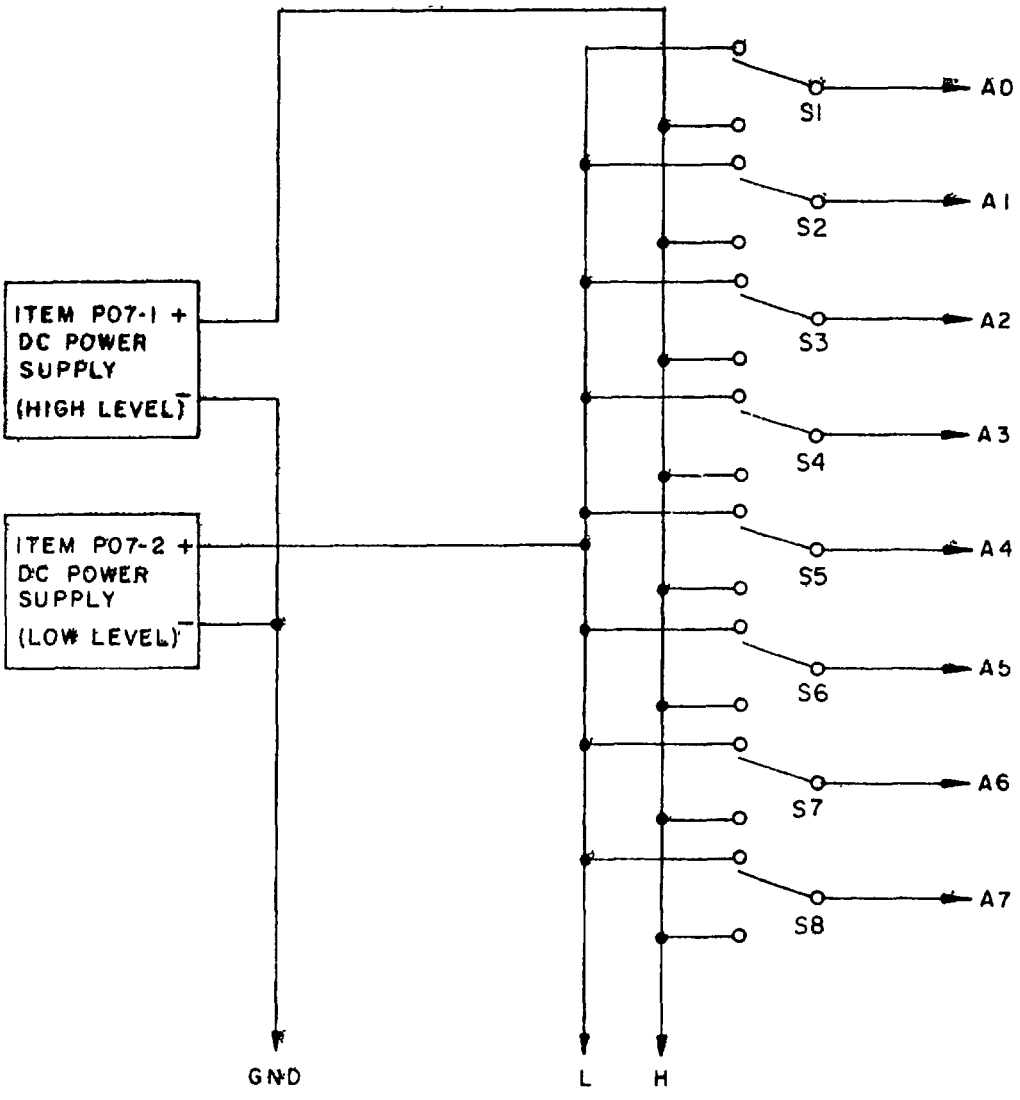
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NOTES:
1. R1 = 499Ω ±1%, 1/4 W.
2. After point A has been connected to the appropriate module pin, the output voltage shall be monitored for a minimum of 5 seconds before the measurement is recorded.
3. Select address with static address generator (either randomly or by referring to the individual firmware pattern) such that the output under test is a low level before connecting point A. If an output is high for all addresses, omit this test for that output.
4. If tests are successively performed, a time interval of 10 seconds minimum between tests must be allowed to preclude detrimental effects due to module heating.

FIGURE 11. V_{OUT} (low-loaded) test circuit for table III.

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NOTES:
1. High level (H) is defined as $2.0\ ^{+0.1}_{-0.0}$ V.
2. Low level (L) is defined as $0.8\ ^{+0.0}_{-0.1}$ V.
3. S1-S8 = SPDT switch.

FIGURE 12. Static address generator.

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TABLE IV. Input breakdown current test.

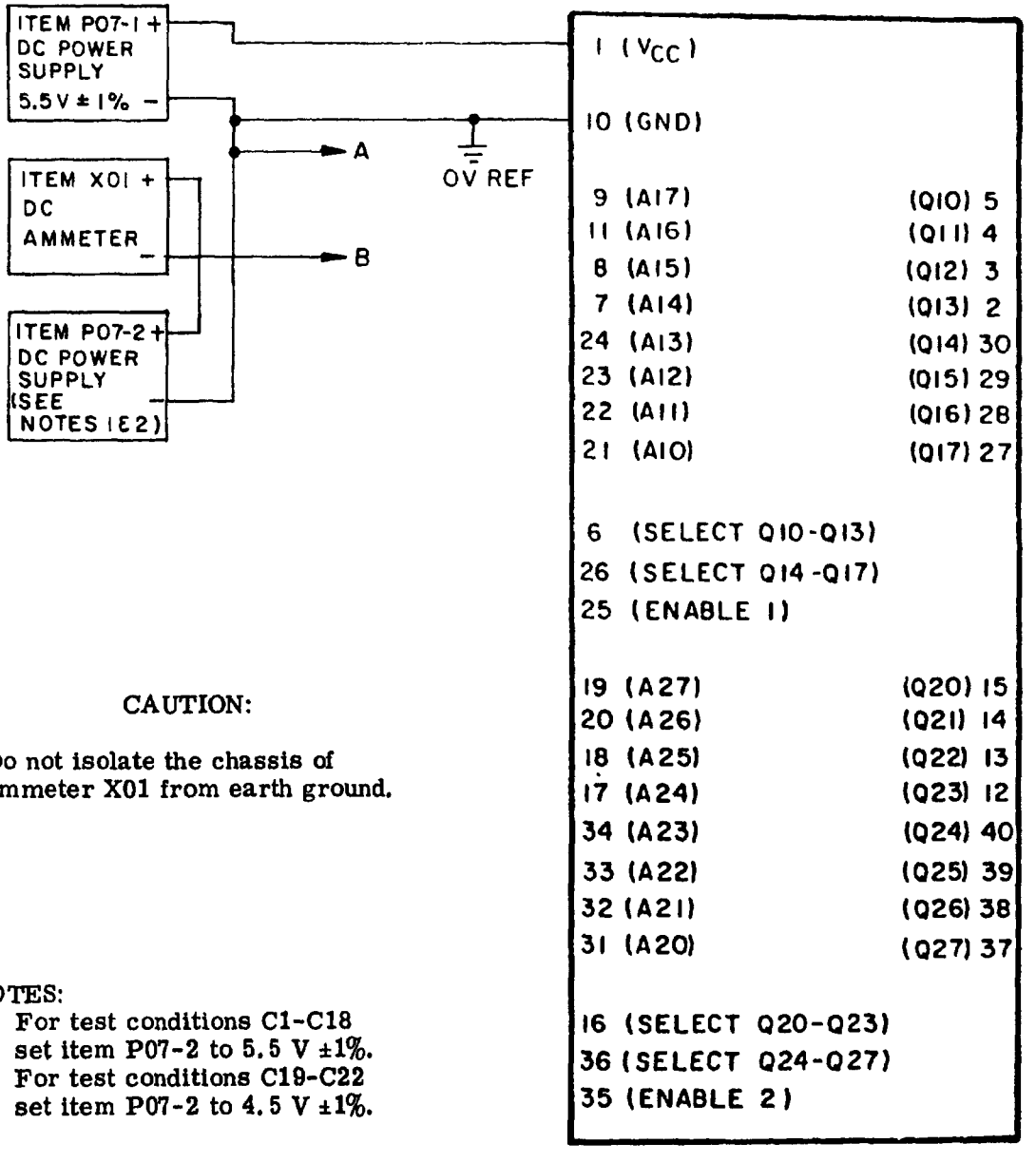
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A17 Input breakdown current	11,8,7,24,23,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	9	0	2	mA	13
C2	A16	9,8,7,24,23,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	11				
C3	A15	9,11,7,24,23,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	8				
C4	A14	9,11,8,24,23,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	7				
C5	A13	9,11,8,7,23,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	24				
C6	A12	9,11,8,7,24,22,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	23				
C7	A11	9,11,8,7,24,23,21,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	22				
C8	A10	9,11,8,7,24,23,22,19,20,18,17,34,33, 32,31,25,35,6,26,16,36	21				
C9	A27	9,11,8,7,24,23,22,21,20,18,17,34,33, 32,31,25,35,6,26,16,36	19				
C10	A26	9,11,8,7,24,23,22,21,19,18,17,34,33, 32,31,25,35,6,26,16,36	20				
C11	A25	9,11,8,7,24,23,22,21,19,20,17,34,33, 32,31,25,35,6,26,16,36	18				
C12	A24	9,11,8,7,24,23,22,21,19,20,18,34,33, 32,31,25,35,6,26,16,36	17				
C13	A23	9,11,8,7,24,23,22,21,19,20,18,17,33, 32,31,25,35,6,26,16,36	34				
C14	A22	9,11,8,7,24,23,22,21,19,20,18,17,34, 32,31,25,35,6,26,16,36	33				
C15	A21	9,11,8,7,24,23,22,21,19,20,18,17,34, 33,31,25,35,6,26,16,36	32				

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C16	A20	Input breakdown current	9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,25,35,6,26,16,36	31	0	2	mA	13
C17	Enable 1		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,35,6,26,16,36	25		2		
C18	Enable 2		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,25,6,26,16,36	35		2		
C19	Select (Q10-Q13)		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,25,35,26,16,36	6		1		
C20	Select (Q14-Q17)		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,25,35,6,16,36	26				
C21	Select (Q20-Q23)		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,25,35,6,26,36	16				
C22	Select (Q24-Q27)		9,11,8,7,24,23,22,21,19,20,18,17,34, 33,32,31,25,35,6,26,16	36				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. For test conditions C1-C18 set item P07-2 to 5.5 V ±1%.
2. For test conditions C19-C22 set item P07-2 to 4.5 V ±1%.

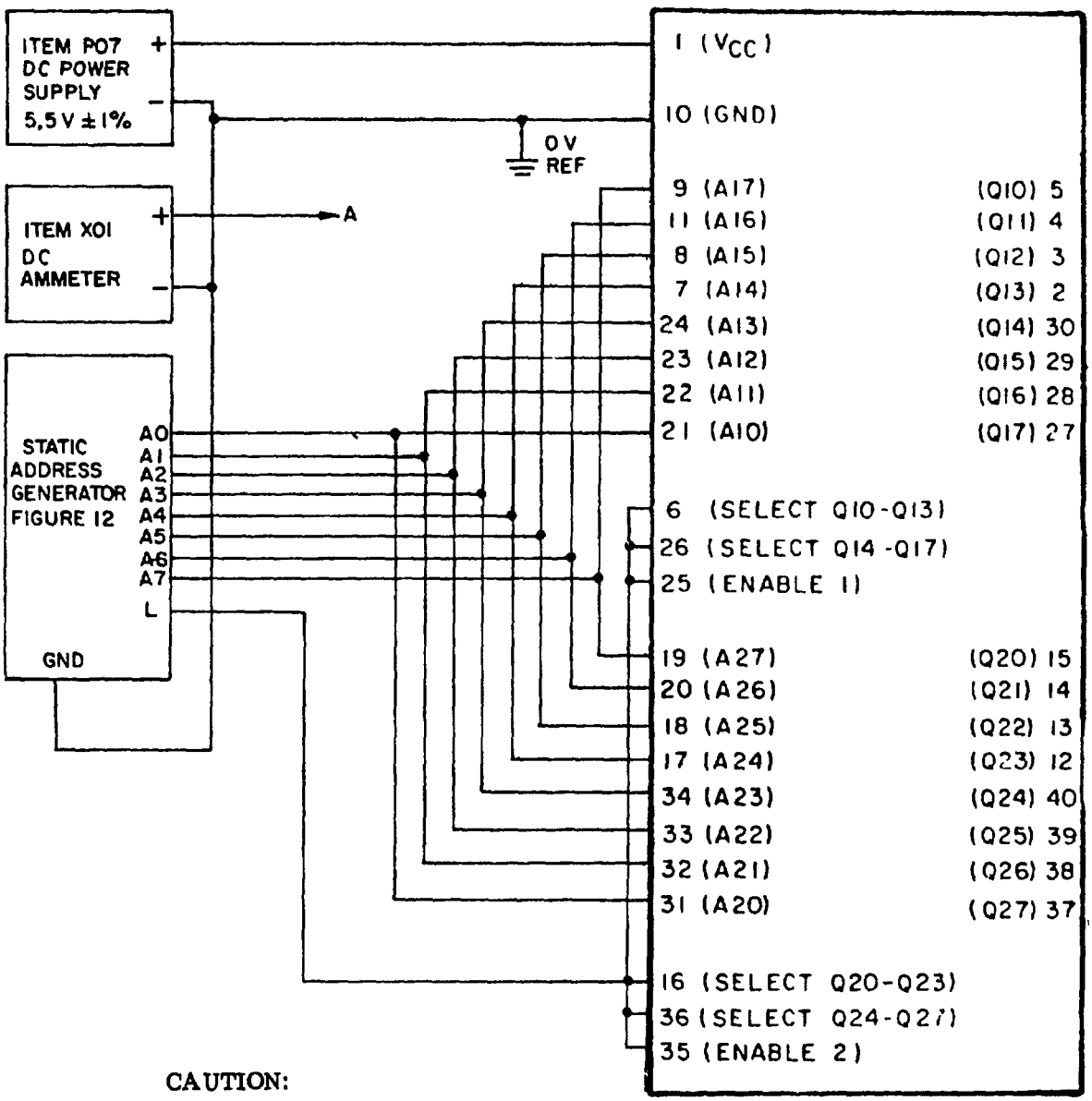
FIGURE 13. Input breakdown current test circuit for table IV.

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TABLE V. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
D1	Q10 Output short circuit current	5	-15	-90	mA	14
D2	Q11	4				
D3	Q12	3				
D4	Q13	2				
D5	Q14	30				
D6	Q15	29				
D7	Q16	28				
D8	Q17	27				
D9	Q20	15				
D10	Q21	14				
D11	Q22	13				
D12	Q23	12				
D13	Q24	40				
D14	Q25	39				
D15	Q26	38				
D16	Q27	37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: Select address with static address generator (either randomly or by referring to the individual firmware pattern) such that the output under test is a high level before connecting point A. If an output is low for all addresses omit this test for that output.

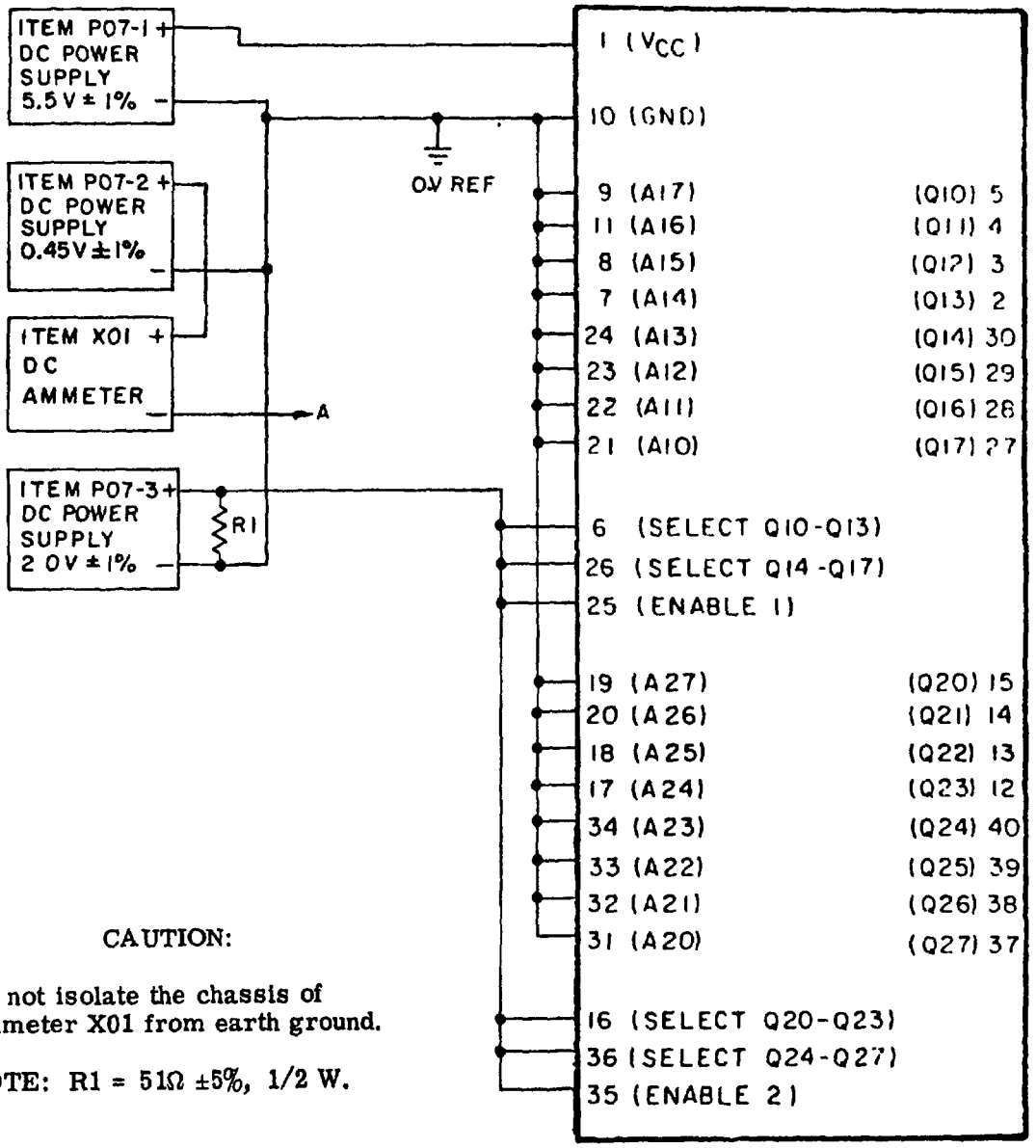
FIGURE 14. Output short circuit current test circuit for table V.

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TABLE VI. Output leakage current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
E1	Q10 Output leakage current, low level	5	-100	100	µA	15
E2	Q11	4				
E3	Q12	3				
E4	Q13	2				
E5	Q14	30				
E6	Q15	29				
E7	Q16	28				
E8	Q17	27				
E9	Q20	15				
E10	Q21	14				
E11	Q22	13				
E12	Q23	12				
E13	Q24	40				
E14	Q25	39				
E15	Q26	38				
E16	Q27	37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 51Ω ±5%, 1/2 W.

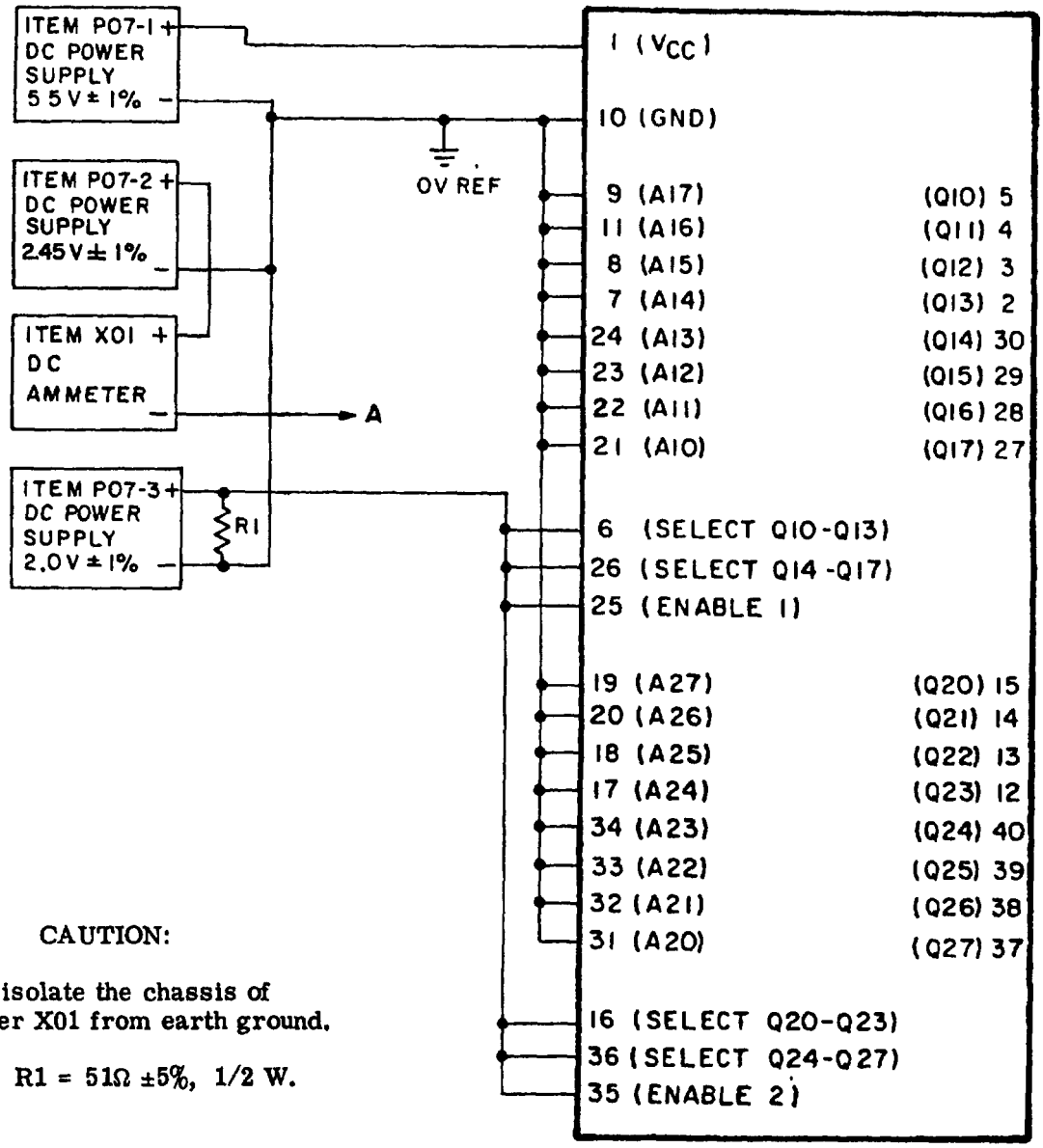
FIGURE 15. Output leakage current (low-level) test circuit for table VI.

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TABLE VII. Output leakage current (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	Q10 Output leakage current, high level	5	-100	100	µA	16
F2	Q11	4				
F3	Q12	3				
F4	Q13	2				
F5	Q14	30				
F6	Q15	29				
F7	Q16	28				
F8	Q17	27				
F9	Q20	15				
F10	Q21	14				
F11	Q22	13				
F12	Q23	12				
F13	Q24	40				
F14	Q25	39				
F15	Q26	38				
F16	Q27	37				

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTE: R1 = 51Ω ±5%, 1/2 W.

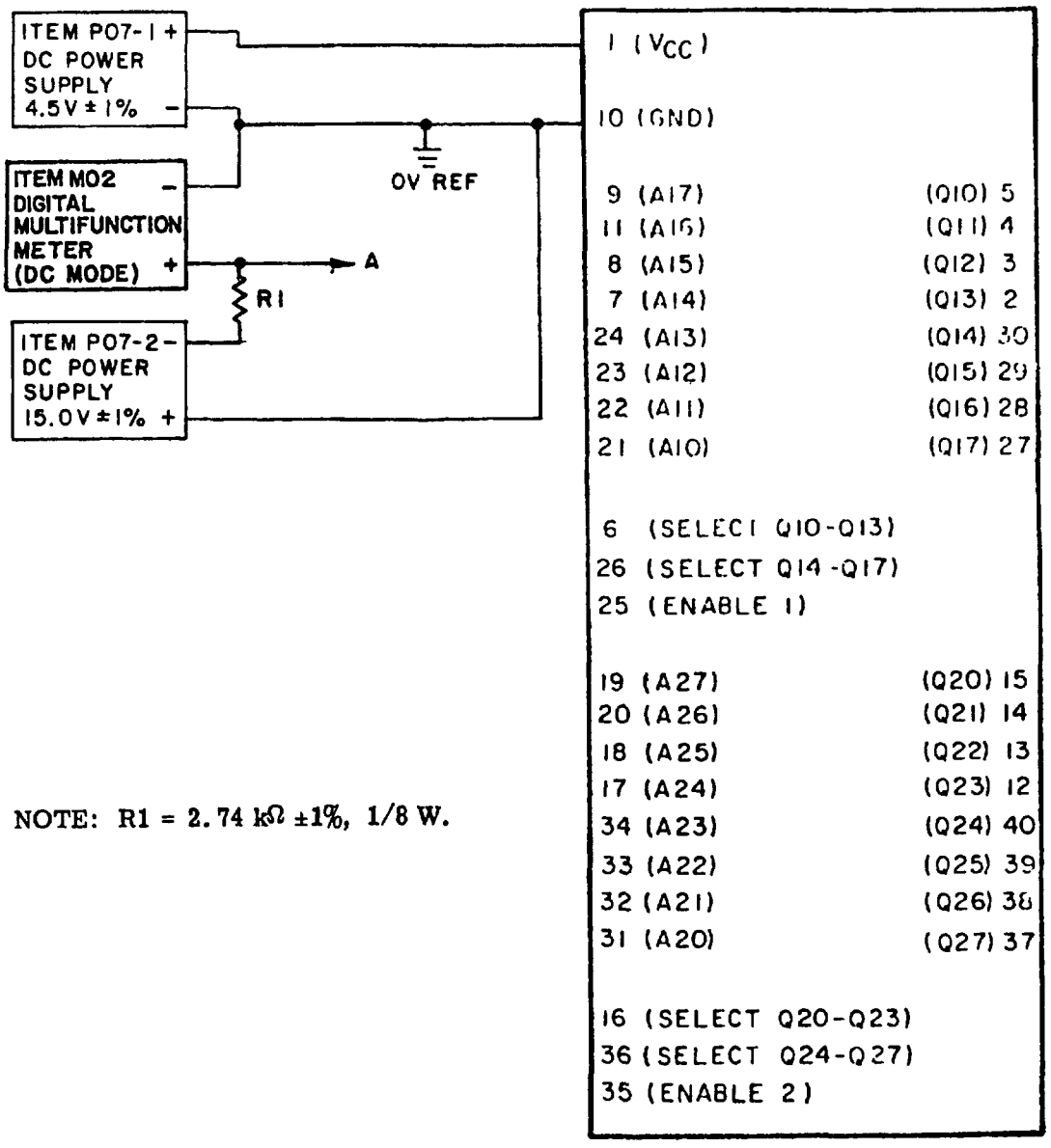
FIGURE 16. Output leakage current (high-level) test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
G1	A17	Input clamp diode voltage	9	-0.4	-1.5	V	17
G2	A16		11				
G3	A15		8				
G4	A14		7				
G5	A13		24				
G6	A12		23				
G7	A11		22				
G8	A10		21				
G9	A27		19				
G10	A26		20				
G11	A25		18				
G12	A24		17				
G13	A23		34				
G14	A22		33				
G15	A21		32				
G16	A20		31				
G17	Select Q10-Q13		6				
G18	Select Q14-Q17		26				
G19	Enable 1		25				
G20	Select Q20-Q23		16				
G21	Select Q24-Q27		36				
G22	Enable 2		35				

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NOTE: R1 = 2.74 kΩ ±1%, 1/8 W.

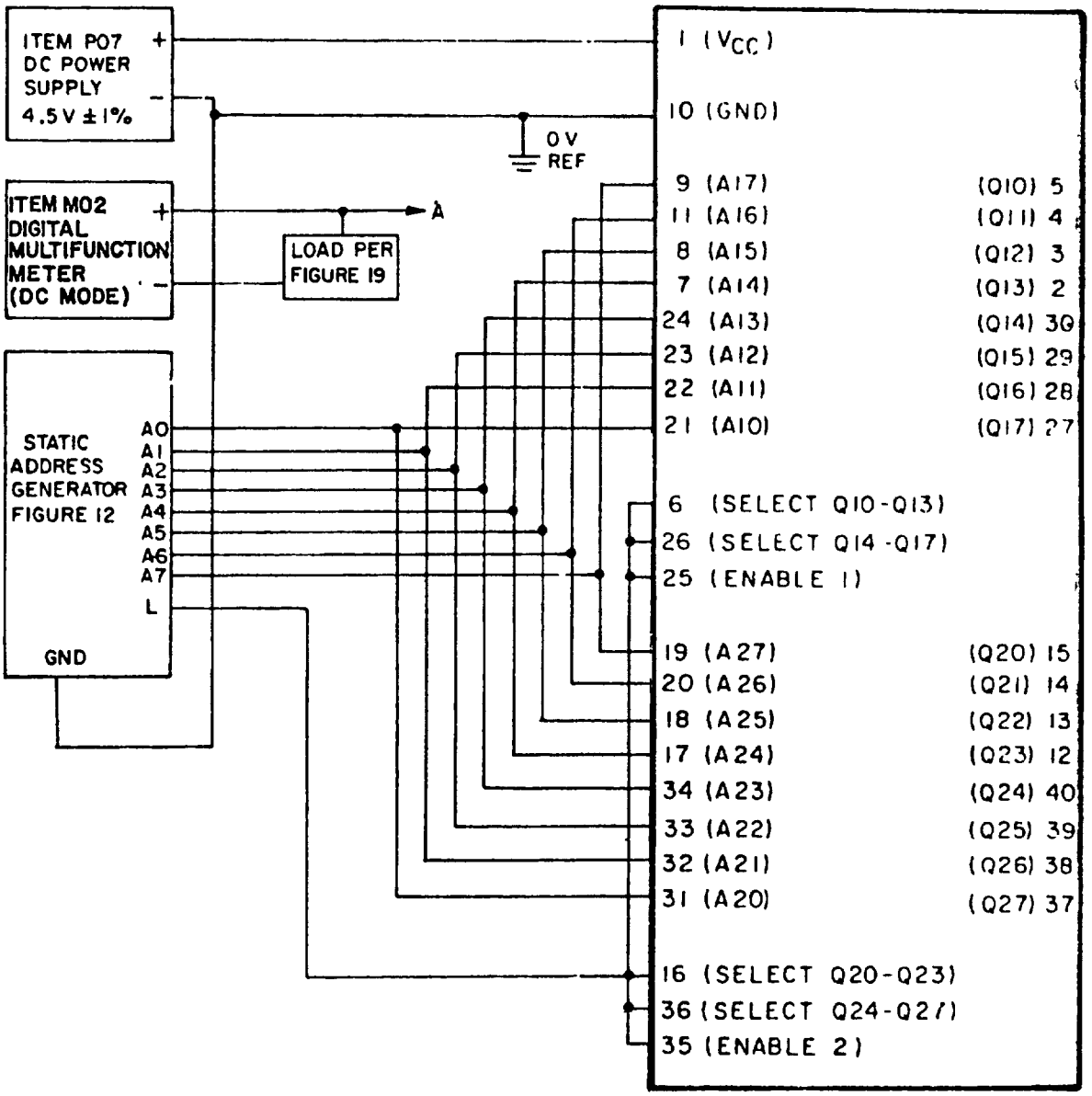
* FIGURE 17. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Static output voltage (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
H1	Q10 Static output voltage, low level	5	0.00	0.45	V	18
H2	Q11	4				
H3	Q12	3				
H4	Q13	2				
H5	Q14	30				
H6	Q15	29				
H7	Q16	28				
H8	Q17	27				
H9	Q20	15				
H10	Q21	14				
H11	Q22	13				
H12	Q23	12				
H13	Q24	40				
H14	Q25	39				
H15	Q26	38				
H16	Q27	37				

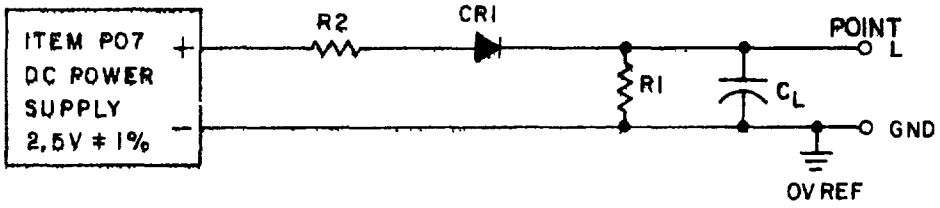
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NOTE: Select address with static address generator (either randomly or by referring to the individual firmware pattern) such that the output under test is a low level. If an output is high for all addresses omit this test for that output.

FIGURE 18. Static output voltage (low-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = 2.61 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 10 mA ±5% out of point "L" with point "L" held at 0.45 V.
4. For test conditions N1-N16, N33-N48, Q1, Q2, R1-R6, and S1-S6, C_L = 50 ±5 pF (including probe and fixture capacitance).
5. For test conditions N17-N32, N49-N64, Q3, and Q4, C_L = 150 ±5 pF (including probe and fixture capacitance).
6. C_L is not required for electrical test conditions H1-H16 and J1-J16 and for functional test condition T.
7. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).
8. For test conditions P1-P64, $C_L \leq 10$ pF (including probe and fixture capacitance).
9. During output switching the 2.5 V power supply shall have a maximum overshoot of 500 mV.

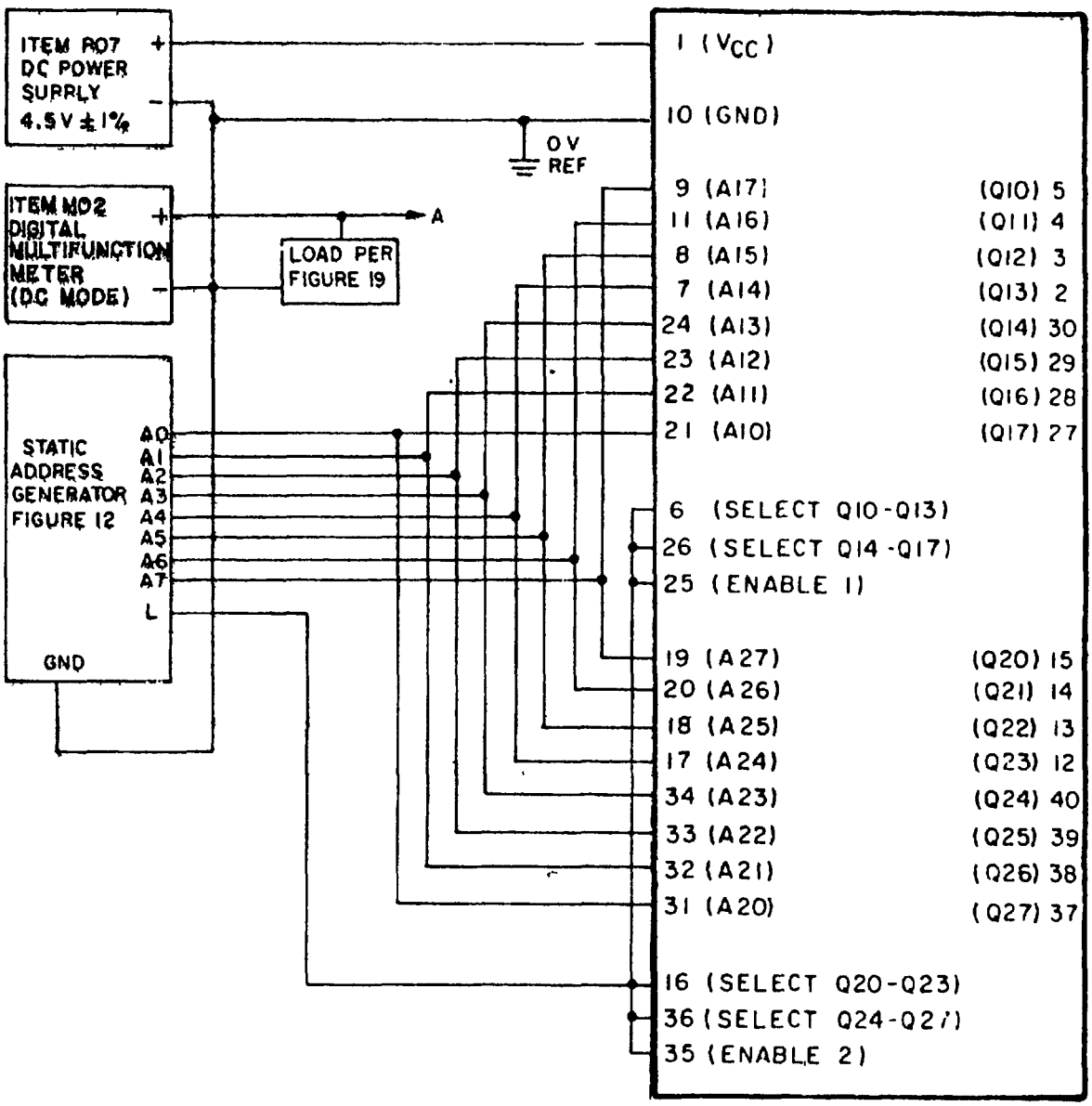
FIGURE 19. Logic load circuit.

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TABLE X. Static output voltage (high-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
J1	Q10 Static output voltage, high level	5	2.4	4.5	V	20
J2	Q11	4				
J3	Q12	3				
J4	Q13	2				
J5	Q14	30				
J6	Q15	29				
J7	Q16	28				
J8	Q17	27				
J9	Q20	15				
J10	Q21	14				
J11	Q22	13				
J12	Q23	12				
J13	Q24	40				
J14	Q25	39				
J15	Q26	38				
J16	Q27	37				

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NOTE: Select address with static address generator (either randomly or by referring to the individual firmware pattern) such that the output under test is a high level. If an output is low for all addresses omit this test for that output.

FIGURE 20. Static output voltage (high-level) test circuit for table X.

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TABLE XI. Input current (low-level and high-level) test.

	Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
*	K1	A17 Input current, low level	Open	9	Open	0	-0.56	mA	21
	K2	A16		11					
	K3	A15		8					
	K4	A14		7					
	K5	A13		24					
	K6	A12		23					
	K7	A11		22					
	K8	A10		21					
	K9	A27		19					
	K10	A26		20					
	K11	A25		18					
	K12	A24		17					
	K13	A23		34					
	K14	A22		33					
	K15	A21		32					
	K16	A20		31					
	K17	Enable 1		25					
	K18	Enable 2		35					
*	K19	Select Q10-Q13		6			-0.28		
	K20	Select Q14-Q17		26					
	K21	Select Q20-Q23		16					
	K22	Select Q24-Q27		36					

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TABLE XI. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	A17 Input current, high level	9	Open	11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36	-20	80	µA	21
L2	A16	11		9,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L3	A15	8		9,11,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L4	A14	7		9,11,8,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L5	A13	24		9,11,8,7,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L6	A12	23		9,11,8,7,24,22,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L7	A11	22		9,11,8,7,24,23,21,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L8	A10	21		9,11,8,7,24,23,22,19,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L9	A27	19		9,11,8,7,24,23,22,21,20,18, 17,34,33,32,31,25,35,6,26, 16,36				
L10	A26	20		9,11,8,7,24,23,22,21,19,18, 17,34,33,32,31,25,35,6,26, 16,36				

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TABLE XI. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L11	A25 Input current, high level	18	Open	9,11,8,7,24,23,22,21,19,20, 17,34,33,32,31,25,35,6,26, 16,36	-20	80	µA	21
L12	A24	17		9,11,8,7,24,23,22,21,19,20, 18,34,33,32,31,25,35,6,26, 16,36				
L13	A23	34		9,11,8,7,24,23,22,21,19,20, 18,17,33,32,31,25,35,6,26, 16,36				
L14	A22	33		9,11,8,7,24,23,22,21,19,20, 18,17,34,32,31,25,35,6,26, 16,36				
L15	A21	32		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,31,25,35,6,26, 16,36				
L16	A20	31		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,25,35,6,26, 16,36				
L17	Enable 1	25		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,35,6,26, 16,36				
L18	Enable 2	35		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,6,26, 16,36				
L19	Select Q10-Q13	6		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,26, 16,36		40		
L20	Select Q14-Q17	26		9,11,8,7,24,23,22,21,19,20, 18,17,34,33,32,31,25,35,6, 16,36		40		

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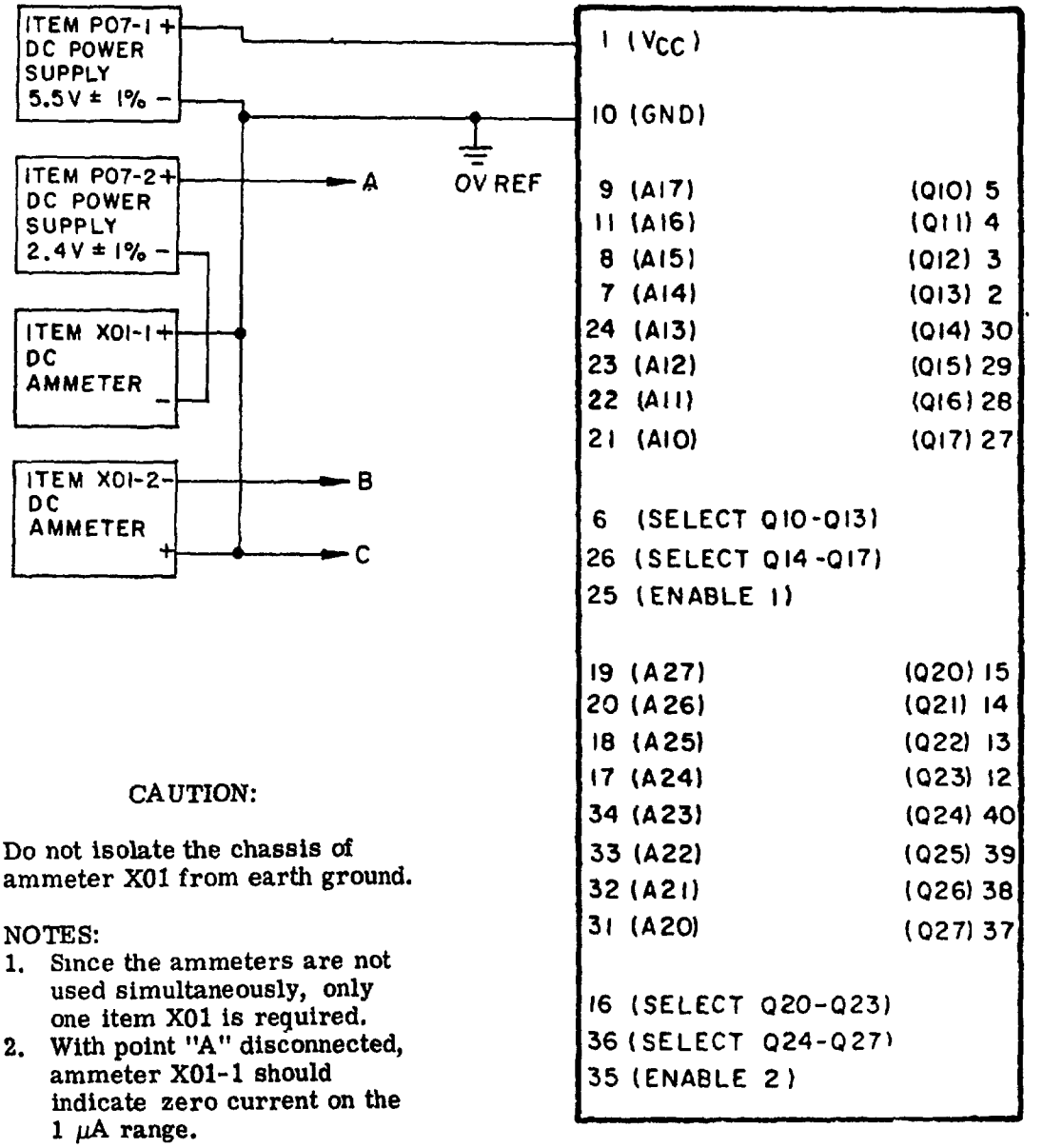
TABLE XI. Input current (low-level and high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L21	Select Q20-Q23 Input current, high level	16	Open	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26,36	-20	40	µA	21
L22	Select Q24-Q27 Input current, high level	36	Open	9,11,8,7,24,23,22,21,19,20,18, 17,34,33,32,31,25,35,6,26,16	-20	40	µA	21

TABLE XII. Power supply current test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	10	450	mA	22

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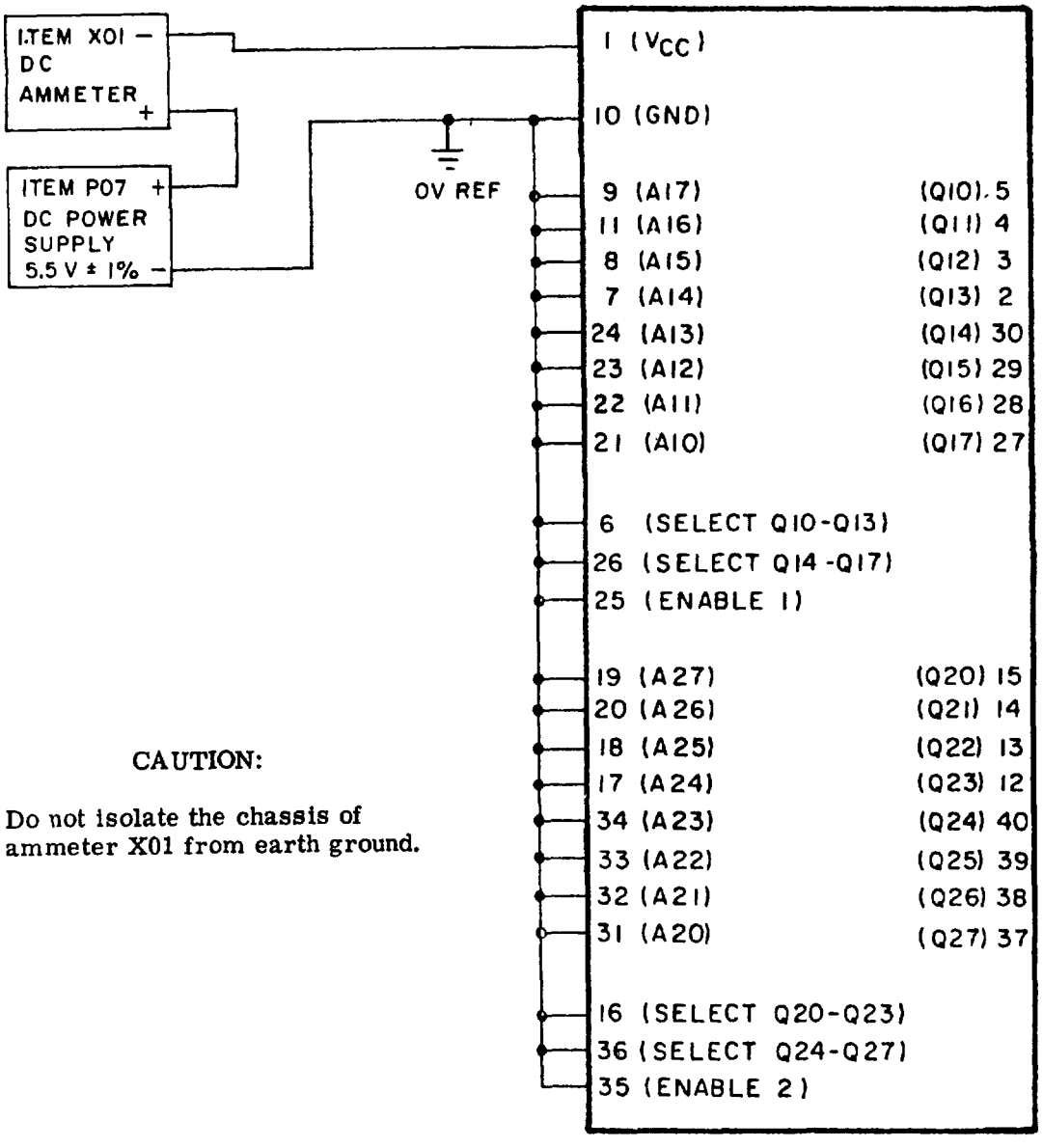
CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

NOTES:
1. Since the ammeters are not used simultaneously, only one item X01 is required.
2. With point "A" disconnected, ammeter X01-1 should indicate zero current on the 1 μA range.

FIGURE 21. Input current (low-level and high-level) test circuit for table XI.

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CAUTION:

Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 22. Power supply current test circuit for table XII.

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
N1	Q10 Transition time (t_{TLH}), C_L = 50 pF	See notes 2 and 3, figure 24	5	5	30	ns	23,24
N2	Q11		4				
N3	Q12		3				
N4	Q13		2				
N5	Q14		30				
N6	Q15		29				
N7	Q16		28				
N8	Q17		27				
N9	Q20		15				
N10	Q21		14				
N11	Q22		13				
N12	Q23		12				
N13	Q24		40				
N14	Q25		39				
N15	Q26		38				
N16	Q27		37				
N17	Q10 Transition time (t_{TLH}), C_L = 150 pF		5		40		
N18	Q11		4				
N19	Q12		3				
N20	Q13		2				
N21	Q14		30				
N22	Q15		29				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

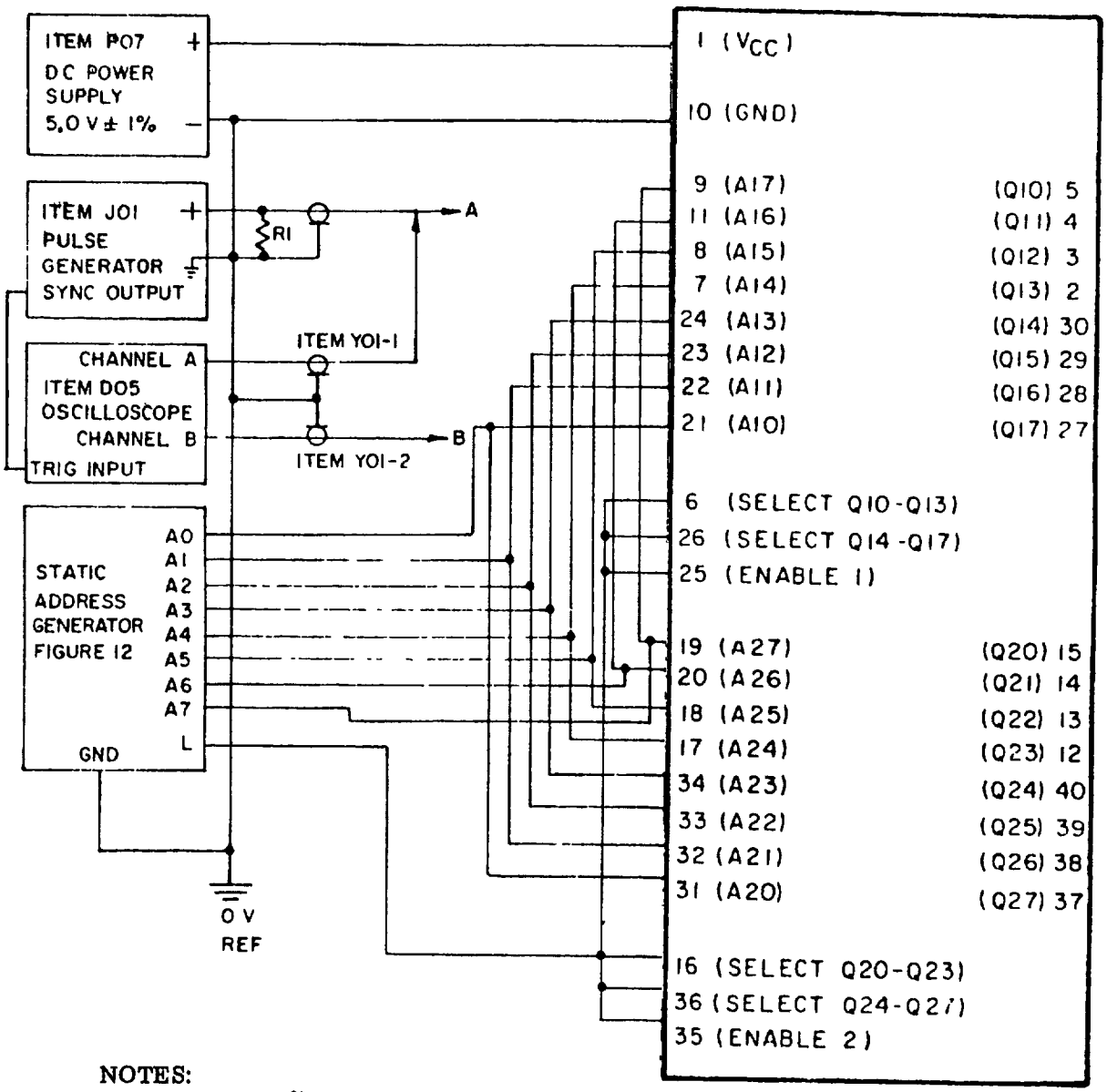
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
N23	Q16 Transition time (t_{TLH}), C_L = 150 pF	See notes 2 and 3, figure 24	28	5	40	ns	23,24
N24	Q17		27				
N25	Q20		15				
N26	Q21		14				
N27	Q22		13				
N28	Q23		12				
N29	Q24		40				
N30	Q25		39				
N31	Q26		38				
N32	Q27		37				
N33	Q10 Transition time (t_{THL}), C_L = 50 pF		5	3	30		
N34	Q11		4				
N35	Q12		3				
N36	Q13		2				
N37	Q14		30				
N38	Q15		29				
N39	Q16		28				
N40	Q17		27				
N41	Q20		15				
N42	Q21		14				
N43	Q22		13				
N44	Q23		12				

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TABLE XIII. Transition time (t_{TLH} and t_{THL}) test - Continued.

	Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
*	N45	Q24 Transition time (t_{THL}), C_L = 50 pF	See notes 2 and 3, figure 24	40	3	30	ns	23,24
	N46	Q25		39				
	N47	Q26		38				
	N48	Q27		37				
	N49	Q10 Transition time (t_{THL}), C_L = 150 pF		5	5	40		
	N50	Q11		4				
	N51	Q12		3				
	N52	Q13		2				
	N53	Q14		30				
	N54	Q15		29				
	N55	Q16		28				
	N56	Q17		27				
	N57	Q20		15				
	N58	Q21		14				
	N59	Q22		13				
	N60	Q23		12				
	N61	Q24		40				
	N62	Q25		39				
	N63	Q26		38				
	N64	Q27		37				

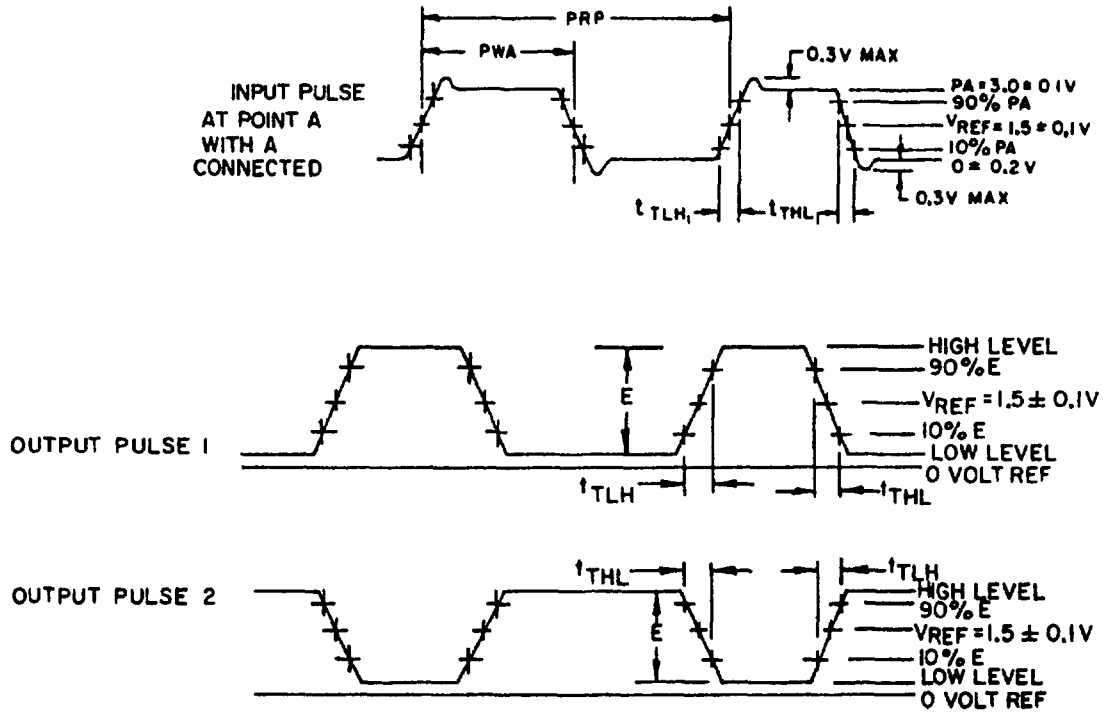
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 19.

FIGURE 23. Transition time (t_{TLH} and t_{THL}) test circuit for table XIII.

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NOTES:

1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 10 \pm 2$ ns.
   PWA = 200 ±50 ns.
   PRP = 400 ±50 ns.
2. Select an address with the static address generator and one address input which will allow (when point A is connected) the output to change from a low level to a high level (output pulse 1 waveform) or a high level to a low level (output pulse 2 waveform). If an output has the same output level for all addresses, omit this test for that output.
3. When point A is connected to an address pin, the corresponding static address generator pin is disconnected.
4. Before making the measurements specified in table XIII, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.45.
5. The lead length of electrical conductors between resistor R1 and input pin and between probes Y01-1 and Y01-2 and input and output pins shall be less than 1 inch (25.4 mm).
6. All voltages are referenced to ground.

FIGURE 24. Transition time (t_{TLH} and t_{THL}) test circuit waveforms for table XIII.

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TABLE XIV. Enable/select recovery time test.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	Q10 Enable recovery time (t_{ERL}), E1	16,36,35,26	25	5	6	5	40	ns	25,26
P2	Q11			4					
P3	Q12			3					
P4	Q13			2					
P5	Q14	16,36,35,6		30	26				
P6	Q15			29					
P7	Q16			28					
P8	Q17			27					
P9	Q20 Enable recovery time (t_{ERL}), E2	6,26,25,36	35	15	16				
P10	Q21			14					
P11	Q22			13					
P12	Q23			12					
P13	Q24	6,26,25,16		40	36				
P14	Q25			39					
P15	Q26			38					
P16	Q27			37					
P17	Q10 Enable recovery time (t_{ERL}), select Q10-Q17	16,36,35,26	6	5	25				
P18	Q11			4					
P19	Q12			3					
P20	Q13			2					

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TABLE XIV. Enable/select recovery time test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P21	Q14 Enable recovery time (t_{ERL}), select Q10-Q17	16,36,35,6	26	30	25	5	40	ns	25,26
P22	Q15			29					
P23	Q16			28					
P24	Q17			27					
P25	Q20 Enable recovery time (t_{ERL}), select Q20-Q27	6,26,25,36	16	15	35				
P26	Q21			14					
P27	Q22			13					
P28	Q23			12					
P29	Q24	6,26,25,16	36	40					
P30	Q25			39					
P31	Q26			38					
P32	Q27			37					
P33	Q10 Enable recovery time (t_{ERH}), E1	16,36,35,26	25	5	6				
P34	Q11			4					
P35	Q12			3					
P36	Q13			2					
P37	Q14	16,36,35,6		30	26				
P38	Q15			29					
P39	Q16			28					
P40	Q17			27					

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TABLE XIV. Enable/select recovery time test - Continued.

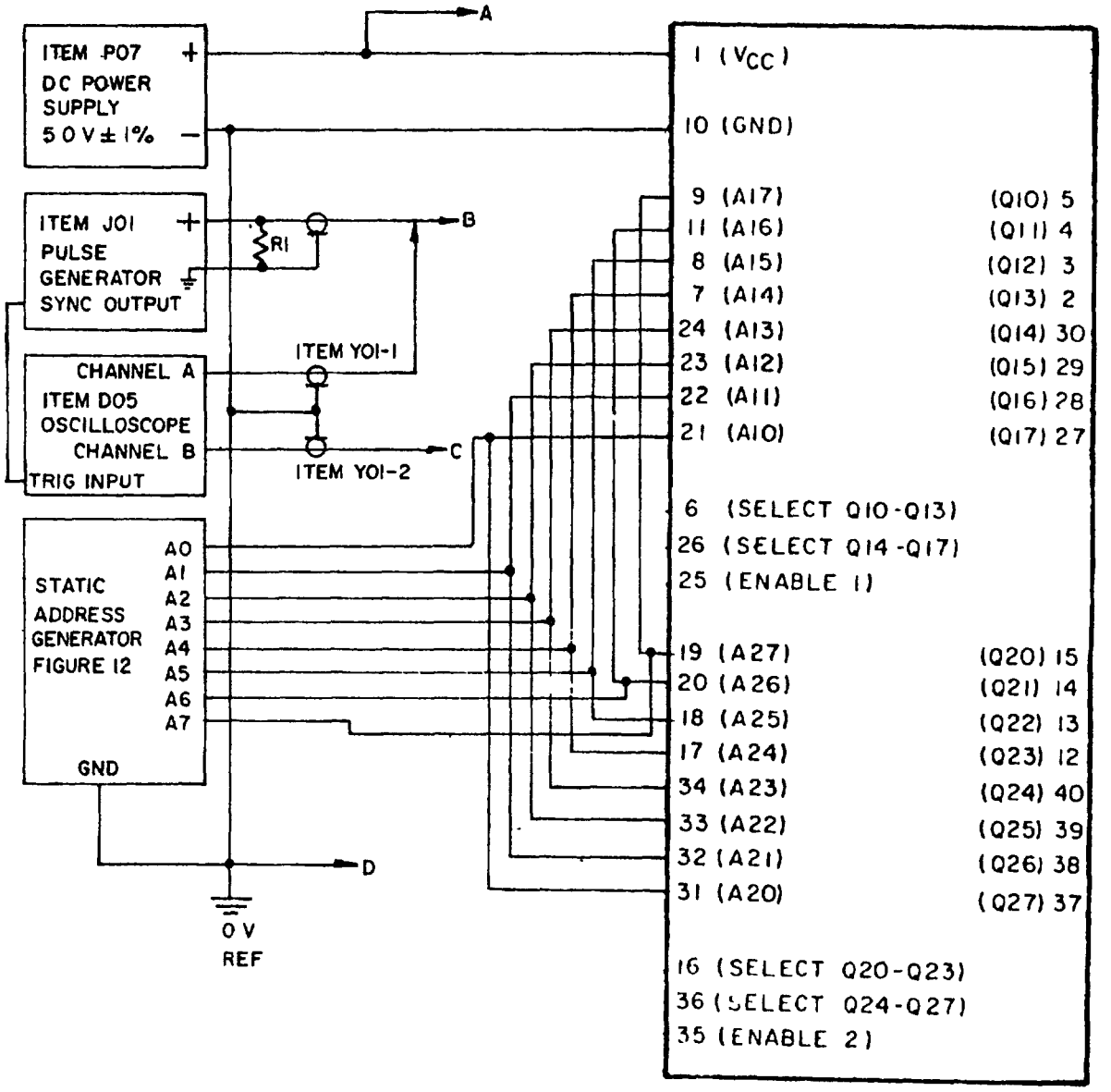
Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P41	Q20 Enable recovery time (t_{ERH}), E2	6,26,25,36	35	15	16	5	40	ns	25,26
P42	Q21			14					
P43	Q22			13					
P44	Q23			12					
P45	Q24	6,26,25,16		40	36				
P46	Q25			39					
P47	Q26			38					
P48	Q27			37					
P49	Q10 Enable recovery time (t_{ERH}), select Q10-Q17	16,36,35,26	6	5	25				
P50	Q11			4					
P51	Q12			3					
P52	Q13			2					
P53	Q14	16,36,35,6	26	30					
P54	Q15			29					
P55	Q16			28					
P56	Q17			27					
P57	Q20 Enable recovery time (t_{ERH}), select Q20-Q27	6,26,25,36	16	15	35				
P58	Q21			14					
P59	Q22			13					
P60	Q23			12					

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TABLE XIV. Enable/select recovery time test - Continued.

Test condition	Parameter	Connect				Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P61	Q24 Enable recovery time (t_{ERH}), select Q20-Q27	6,26,25,16	36	40	35	5	40	ns	25,26
P62	Q25	↓	↓	39	↓	↓	↓	↓	↓
P63	Q26	↓	↓	38	↓	↓	↓	↓	↓
P64	Q27	↓	↓	37	↓	↓	↓	↓	↓

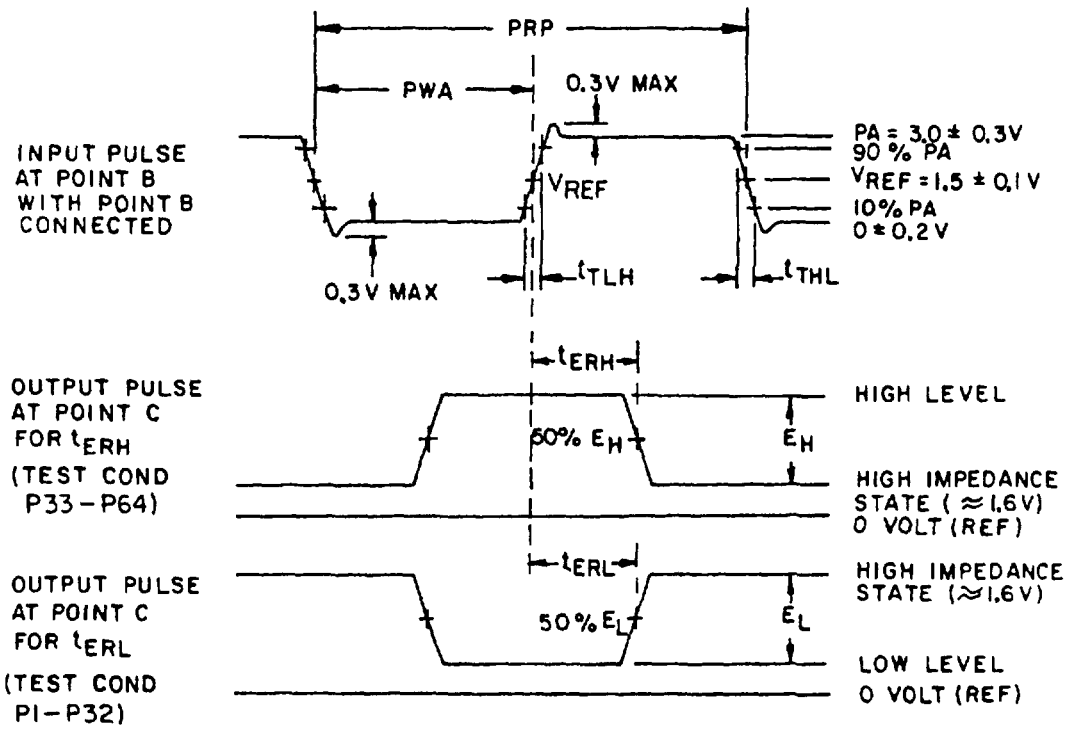
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. Load output under test with a load as specified on figure 19.

FIGURE 25. Enable/select recovery time test circuit for table XIV.

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NOTES:

1. Input pulse parameters:
   PWA = 500 ±50 ns.
   PRP = 1, 000 ±100 ns.
   $t_{TLH} = t_{THL} = 10 \pm 2$ ns.
2. Select an address with the static address generator and one address input which will allow (when point B is connected) the output to change from a low level to the high-impedance level (t_{ERL}) or a high level to the high-impedance level (t_{ERH}). If an output has a high level for all addresses, omit the t_{ERL} test for that output. If an output has a low level for all addresses, omit the t_{ERH} test for that output.
3. When point B is connected to an enable pin, the static address generator output (L) is disconnected.
4. Load each output with the load as specified on figure 19.
5. Before making the measurements as specified in table XIV, verify that the output pulse high level is $\geq$ 2.4 V and the low level is $\leq$ 0.5 V.
6. The lead length of electrical conductors between resistor R1 and input pin and between probes Y01-1 and Y01-2 and input and output pins shall be less than 1 inch (25..4 mm).
7. All voltages are referenced to ground.

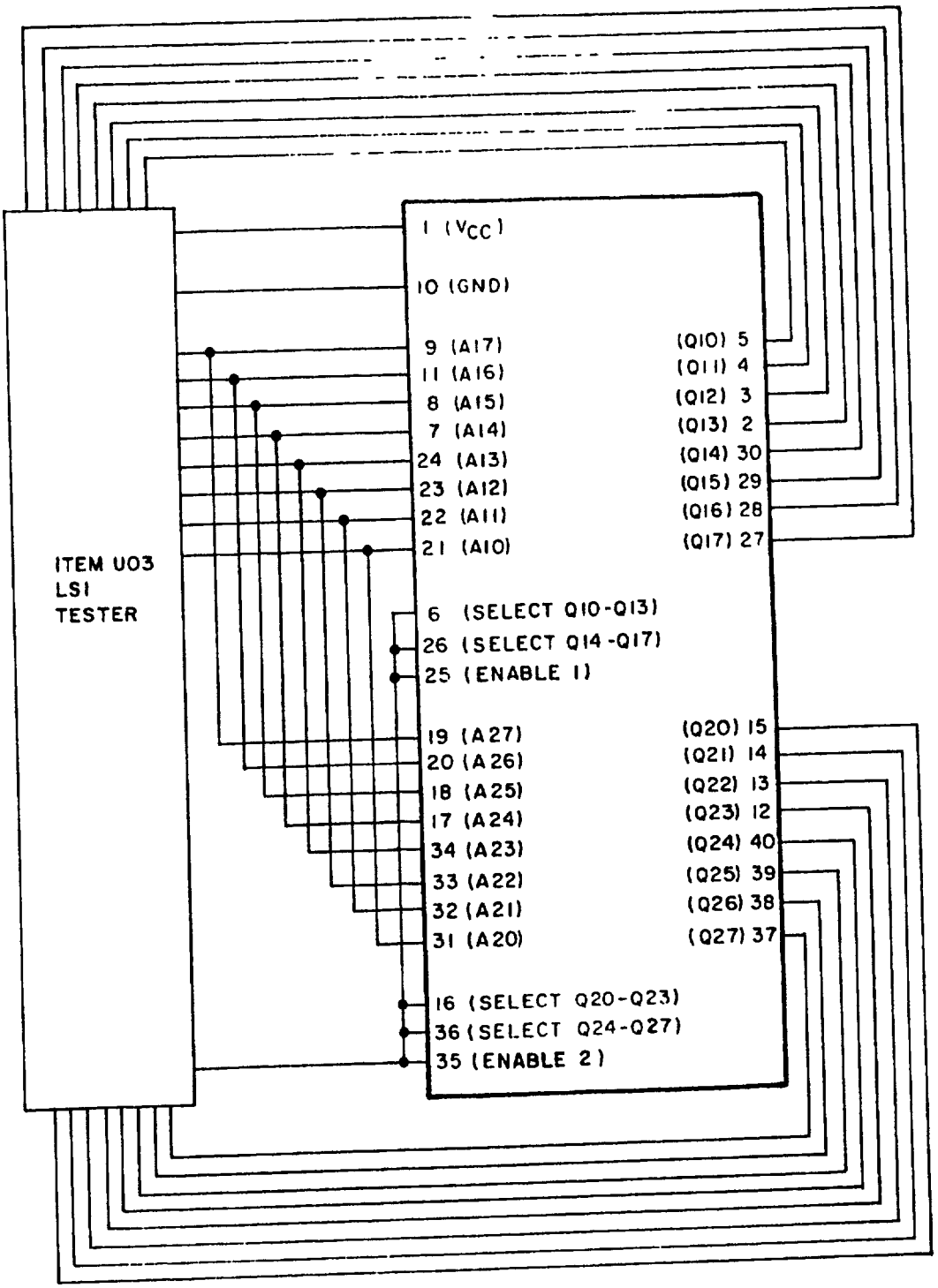
FIGURE 26. Enable/select recovery time test circuit waveforms for table XIV.

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TABLE XV. Galloping read access pattern test.

Test condition	Parameter		Initial limits 25°±5°C ambient		Unit	Test figure
			Min	Max		
Q1	Galloping read access pattern (C_L = 50 pF, V_{CC} = 4.5 V)	Module shall conform to the requirements of figures 27 and 28 for each test condition.	No failure			
Q2	Galloping read access pattern (C_L = 50 pF, V_{CC} = 5.5 V)					
Q3	Galloping read access pattern (C_L = 150 pF, V_{CC} = 4.5 V)					
Q4	Galloping read access pattern (C_L = 150 pF, V_{CC} = 5.5 V)					

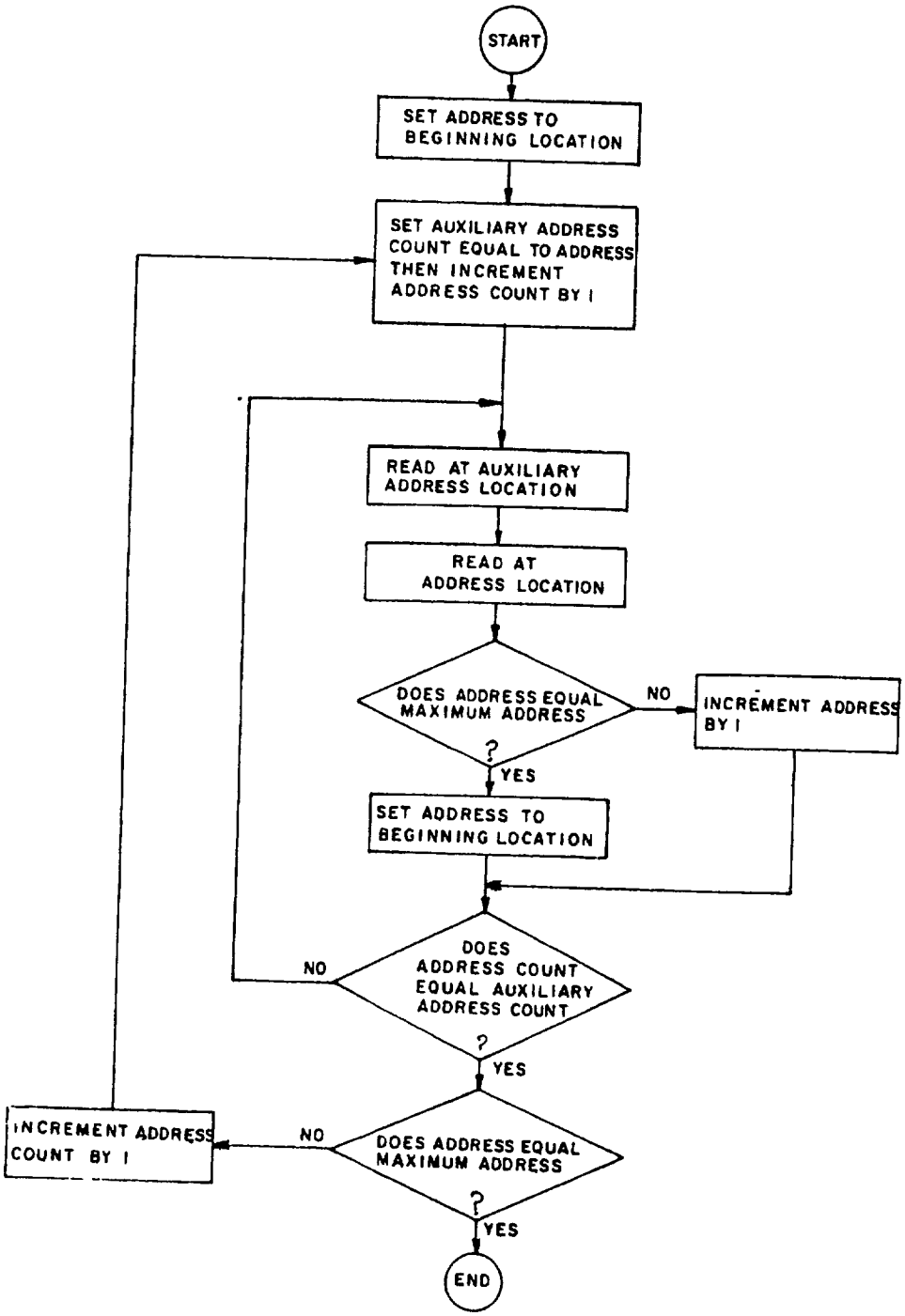
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NOTES:
1. V_{CC} = 4.5 V ±1% for test conditions Q1 and Q3.
2. V_{CC} = 5.5 V ±1% for test conditions Q2 and Q4.
3. Load output under test with a load as specified on figure 19.

FIGURE 27. Galloping read access pattern test circuit for table XV.

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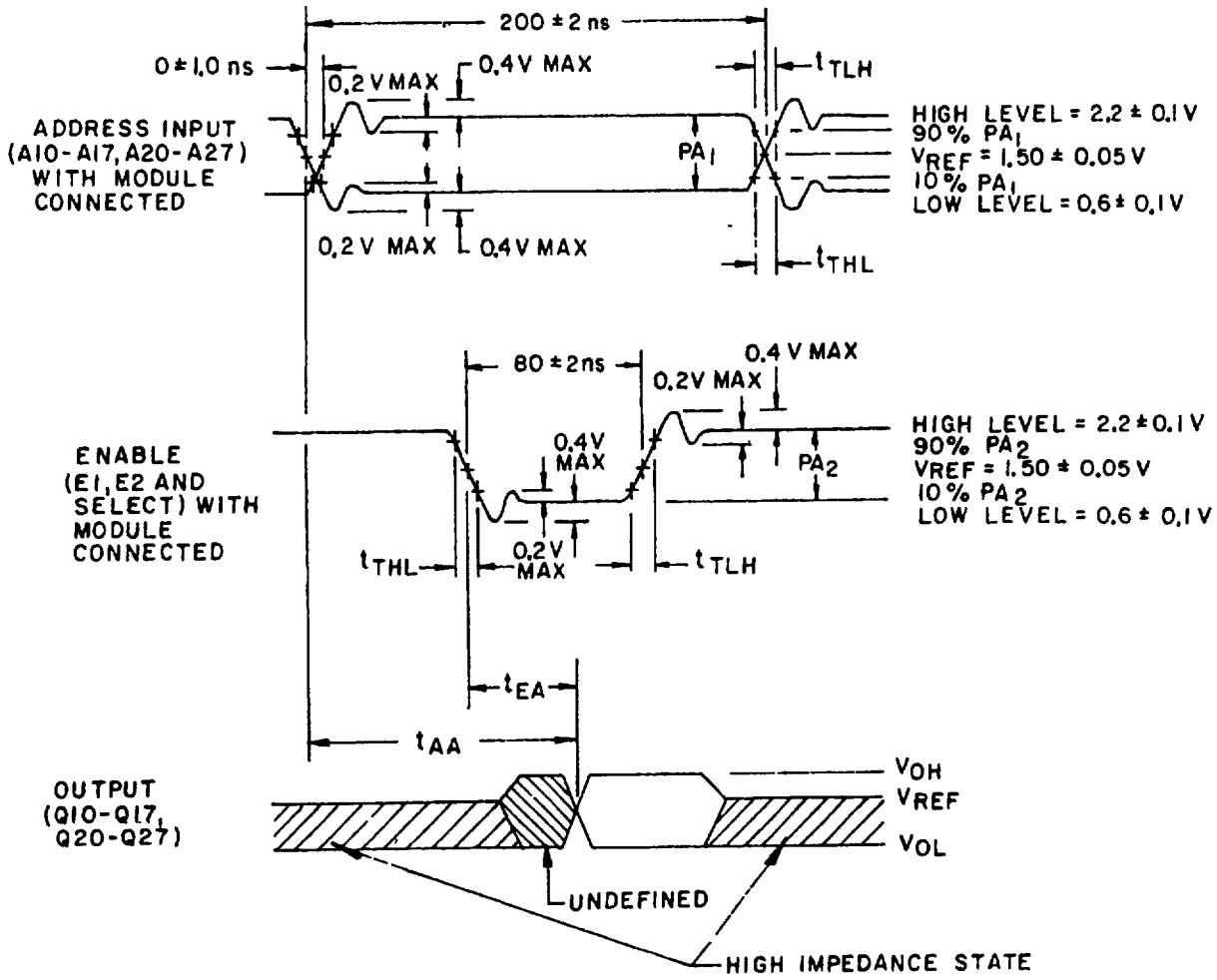


GALLOPING READ ACCESS PATTERN PROGRAM FLOW CHART

NOTE: Pattern is read and verified at each location.

FIGURE 28. Galloping read access pattern program flow chart, waveforms, and program for table XV.

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READ ACCESS INPUT AND OUTPUT WAVEFORMS

NOTES:
1. High-impedance state = 1.6 V.
2. Address access time (50 pF load) $t_{AA1} = 80\ ^{+0}_{-4}$ ns.
3. Address access time (150 pF load) $t_{AA2} = 90\ ^{+0}_{-4}$ ns.
4. Enable access time (50 pF load) $t_{EA1} = 45\ ^{+0}_{-2}$ ns.
5. Enable access time (150 pF load) $t_{EA2} = 55\ ^{+0}_{-2}$ ns.
6. $t_{TLH} = t_{THL} = 10 \pm 4$ ns.

FIGURE 28. Galloping read access pattern program flow chart, waveforms, and program for table XV - Continued.

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Program for test conditions Q1-Q4:

1. The objective of the galloping read access pattern is to test address access time. It is necessary to test all possible address transitions. The test also verifies circuit enable time (t_{EA}) and address access time (t_{AA}).

2. A program that fulfills the galloping read access test objective would be:

a. Read memory address zero.
b. Read memory address one.
c. Read memory address zero.
d. Read memory address two.
e. Read memory address zero.
f. Read memory address three.

g. Read memory address zero.
h. Read memory address 256.
i. Read memory address one.
j. Read memory address two.
k. Read memory address one.
l. Read memory address three.

m. Read memory address one.
n. Read memory address 256.
o. Read memory address one.
p. Read memory address zero.
q. Read memory address two.
r. Read memory address three.
s. Read memory address two.
t. Read memory address four.

u. Continue this pattern until all memory addresses have been tested to all other memory addresses.

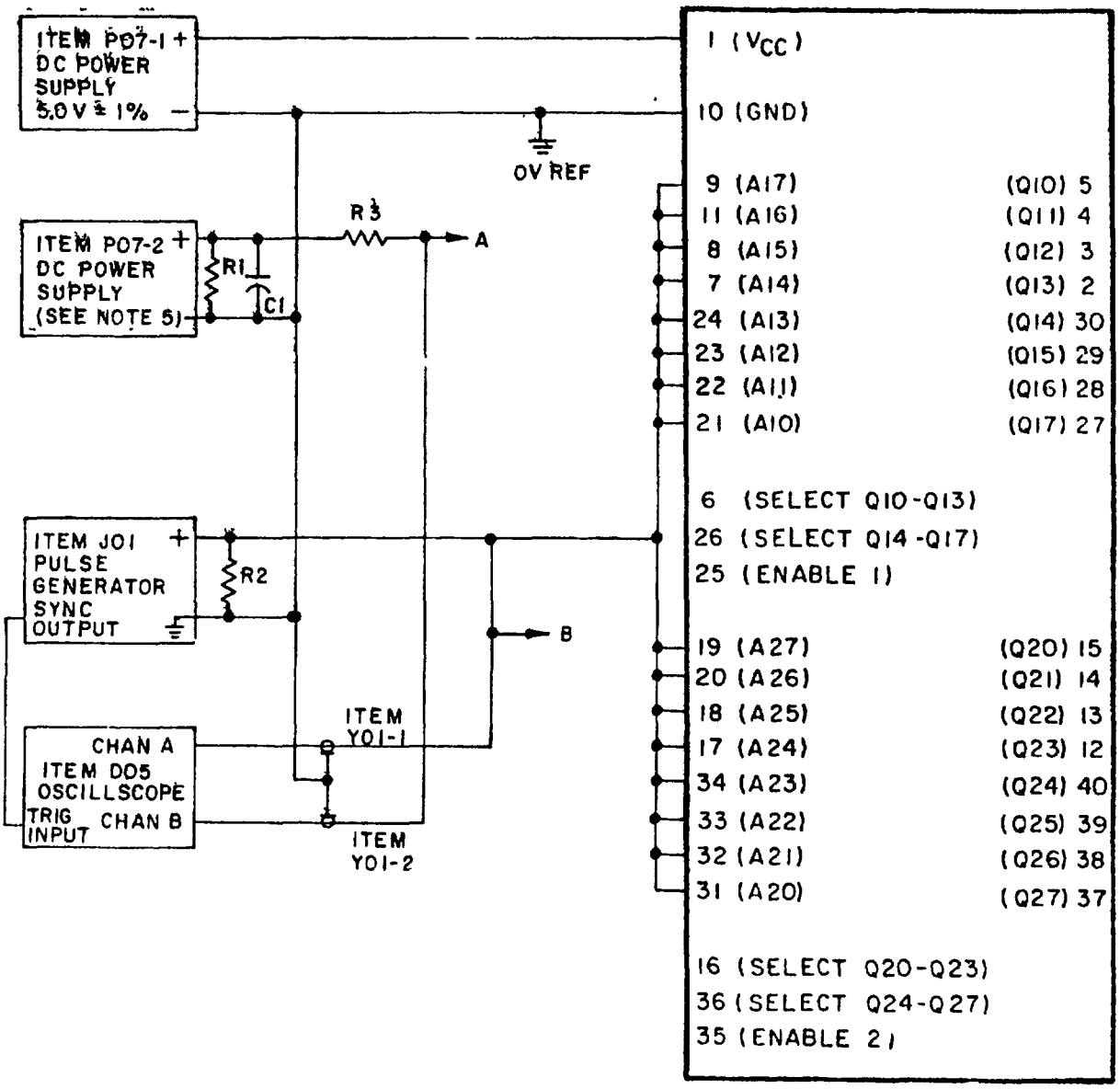
FIGURE 28. Galloping read access pattern program flow chart, waveforms, and program for table XV - Continued.

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TABLE XVI. Input crosstalk (low-level and high-level) test.

Test condition	Parameter		Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
R1	Enable 1	Input crosstalk low level	25	35,6,26,16,36	0	400	mV	29,30
R2	Enable 2		35	25,6,26,16,36				
R3	Select Q10-Q13		6	25,35,26,16,36				
R4	Select Q14-Q17		26	25,35,6,16,36				
R5	Select Q20-Q23		16	25,35,6,26,36				
R6	Select Q24-Q27		36	25,35,6,26,16				
S1	Enable 1	Input crosstalk, high level	25	35,6,26,16,36				
S2	Enable 2		35	25,6,26,16,36				
S3	Select Q10-Q13		6	25,35,26,16,36				
S4	Select Q14-Q17		26	25,35,6,16,36				
S5	Select Q20-Q23		16	25,35,6,26,36				
S6	Select Q24-Q27		36	25,35,6,26,16				

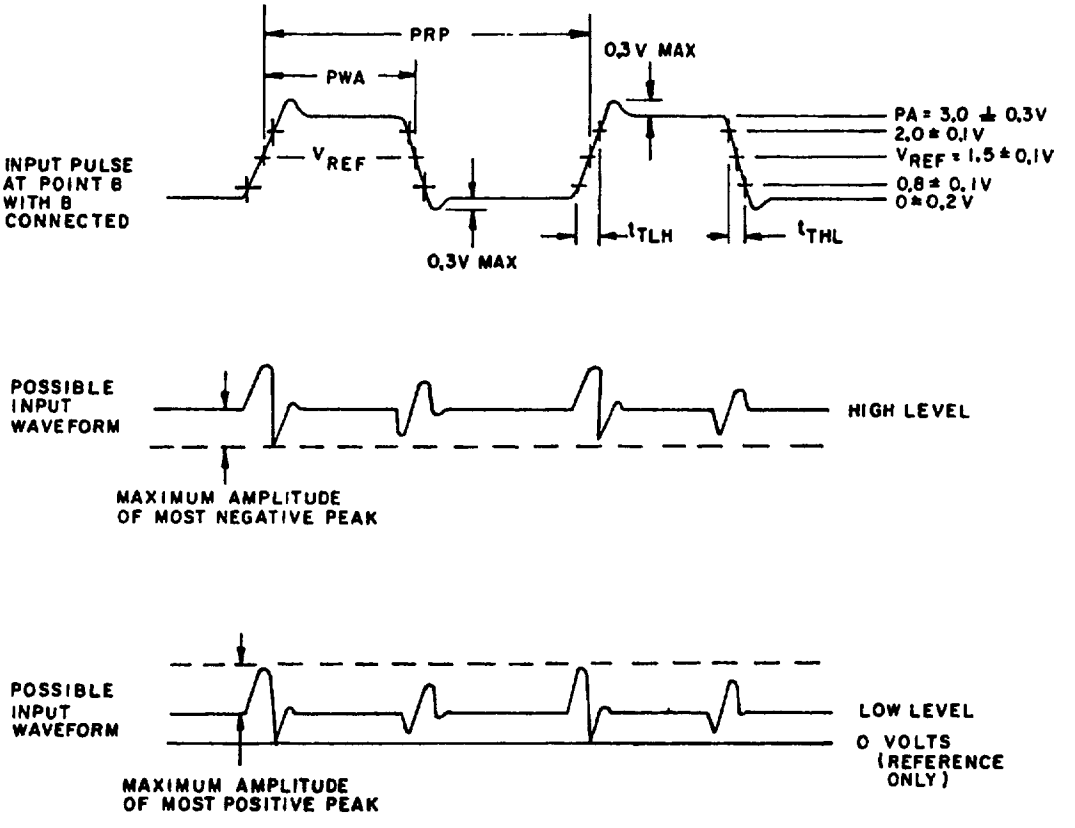
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. R3 = 150Ω ±1%, 1/8 W.
3. C1 = Add sufficient capacitance to reduce item P07-2 power supply noise to < 50 mV.
4. Load all outputs with a load as specified on figure 19.
5. For test conditions R1-R6, set item P07-2 to 0.45 V ±1%. For test conditions S1-S8 set item P07-2 to 2.4 V ±1%.

FIGURE 29. Input crosstalk (low-level and high-level) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns verified at each input.
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors shall be less than 2 inches (50.8 mm) between: resistors and input pins; probes Y01-1 and Y01-2 and input pins; load circuits and output pins; ground leads of probes Y01-1 and Y01-2 and test fixture and ground pins.

FIGURE 30. Input crosstalk (low-level and high-level) test circuit waveforms for table XVI.

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TABLE XVII. Functional test pattern test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
T	Functional test pattern	← See note 1, figure 31 →			

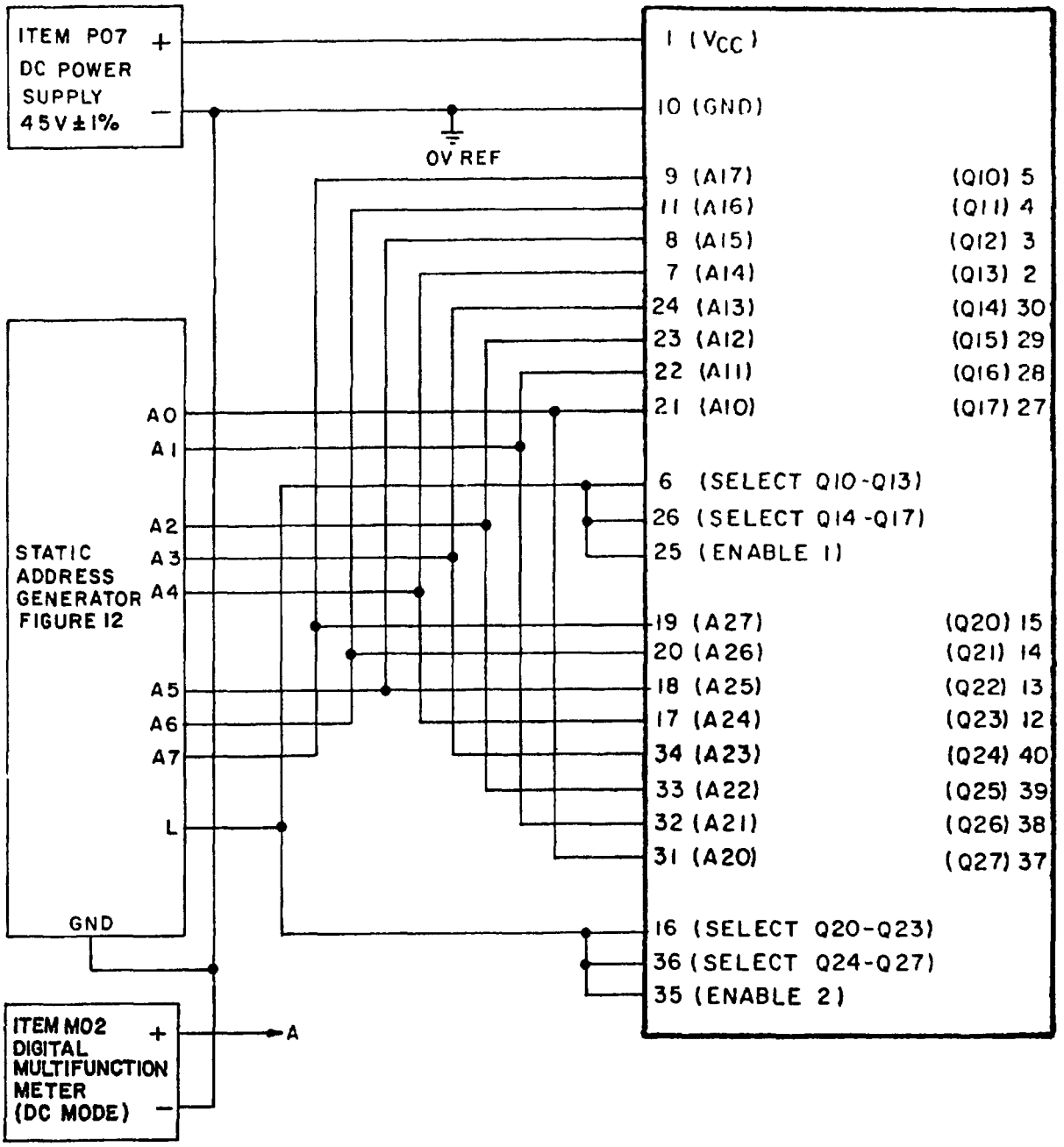
TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	32, 33

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect	Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V	Power supply filter capacitance	1	1.6	3.0	µF	34

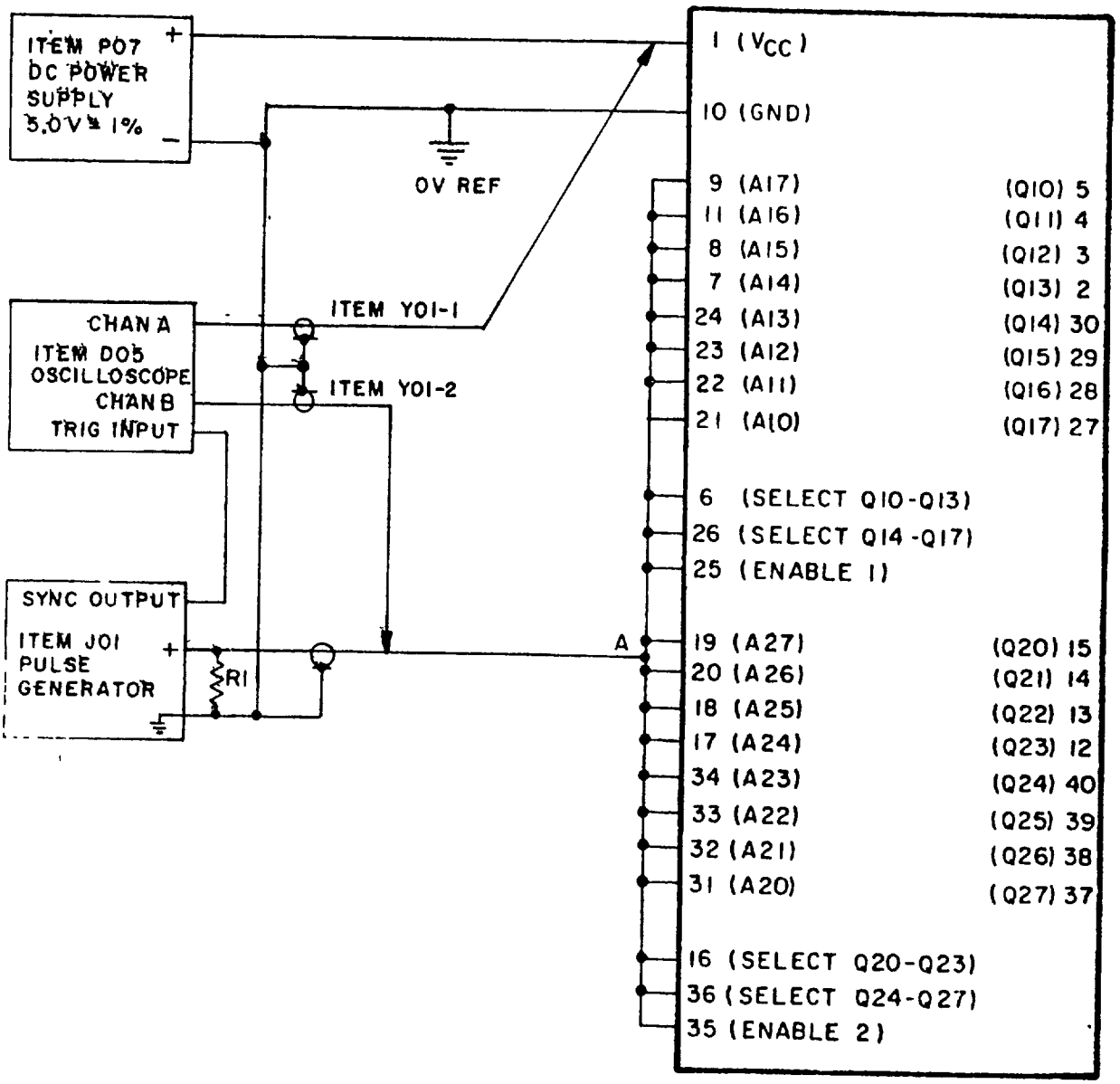
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NOTES:
1. The static address generator is used to select, in turn, each of the 256 address combinations. For each address combination the output is monitored at point "A" to verify the firmware pattern for the module. Output high level shall be $\geq$ 2.4 V and output low level shall be $\leq$ 0.5 V.
2. Load the output under test with the load as specified on figure 19.

FIGURE 31. Functional test pattern test circuit for table XVII.

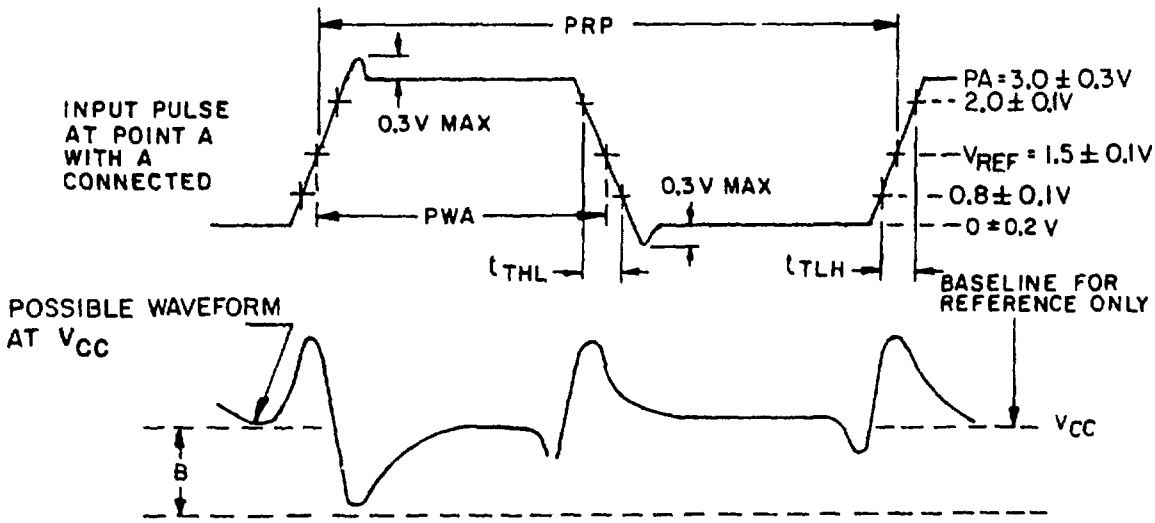
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NOTE: R1 = 51Ω ±5%, 1 W.

FIGURE 32. Power supply transient amplitude test circuit for table XVIII.

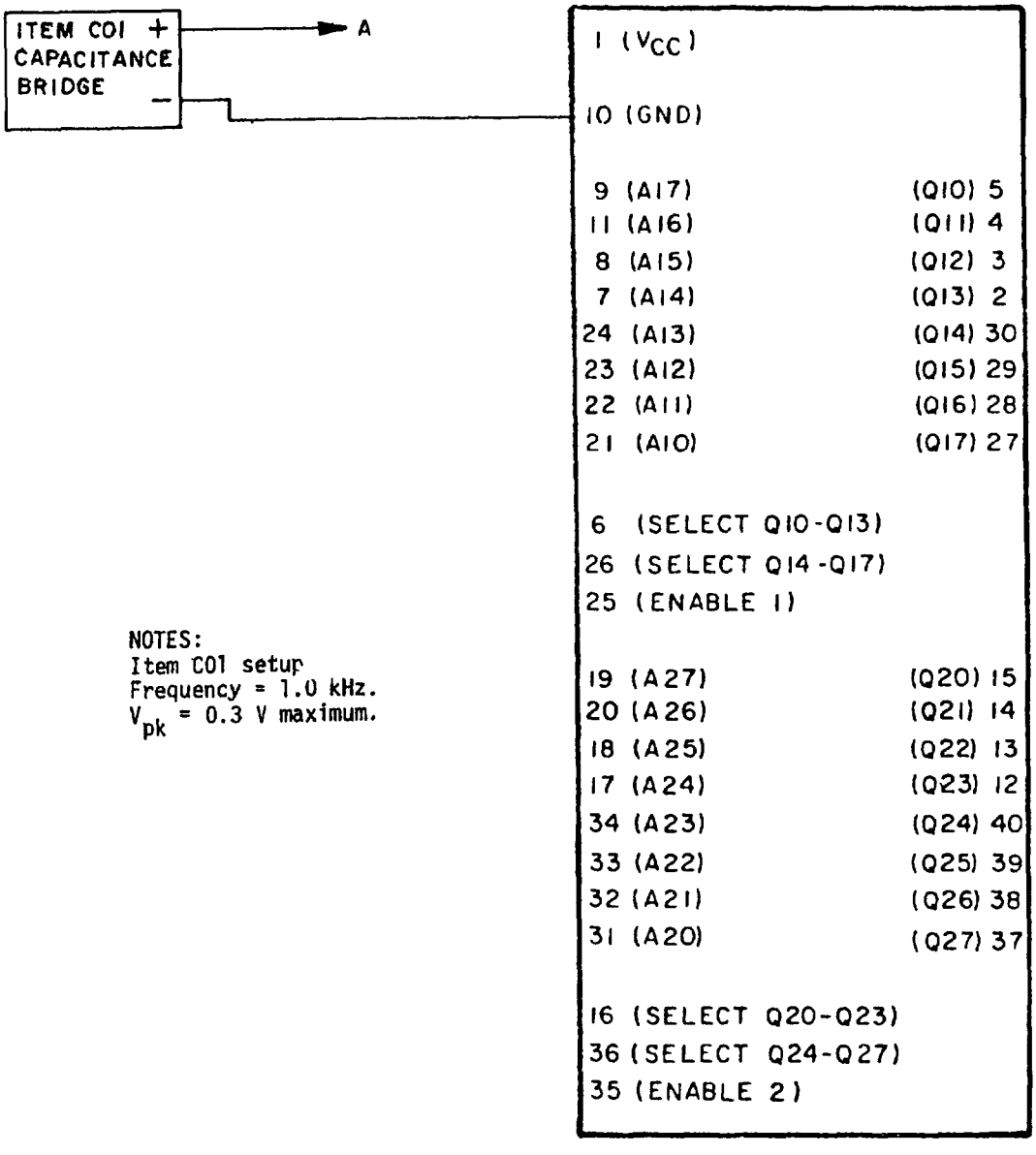
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NOTES:
1. Point A input parameters:
   PWA = 1.0 ±0.1 μs.
   PRP = 2.0 ±0.1 μs.
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input).
2. B = Measurement from the baseline to the most negative excursion of the waveform.
3. All voltages are referenced to ground.

FIGURE 33. Power supply transient amplitude test circuit waveforms for table XVIII.

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* FIGURE 34. Power supply filter capacitance test circuit for table XIX.

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TABLE XX. Input and output capacitance test.

Test condition	Parameter	Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W1	A17	9	2	30	pF	35
W2	A16	11				
W3	A15	8				
W4	A14	7				
W5	A13	24				
W6	A12	23				
W7	A11	22				
W8	A10	21				
W9	A27	19				
W10	A26	20				
W11	A25	18				
W12	A24	17				
W13	A23	34				
W14	A22	33				
W15	A21	32				
W16	A20	31				
W17	Enable 1	25				
W18	Enable 2	35				
W19	Select Q10-Q13	6		20		
W20	Select Q20-Q23	16				
W21	Select Q24-Q27	36				
W22	Select Q14-Q17	26				

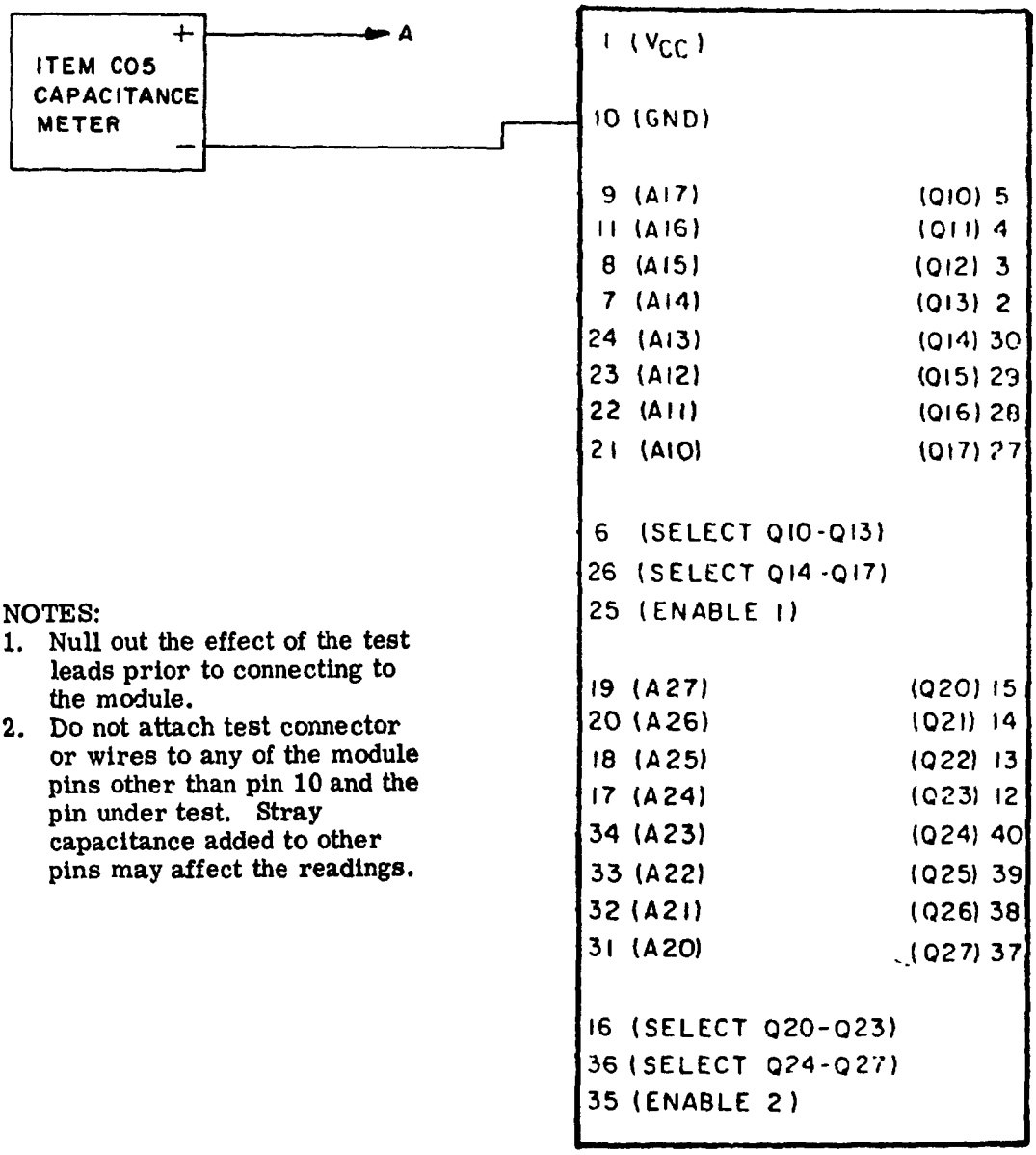
Input capacitance

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TABLE XX. Input and output capacitance test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
W23	Q10	Output capacitance	5	2	15	pF	35
W24	Q11		4				
W25	Q12		3				
W26	Q13		2				
W27	Q14		30				
W28	Q15		29				
W29	Q16		28				
W30	Q17		27				
W31	Q24		40				
W32	Q25		39				
W33	Q26		38				
W34	Q27		37				
W35	Q20		15				
W36	Q21		14				
W37	Q22		13				
W38	Q23		12				

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NOTES:
1. Null out the effect of the test leads prior to connecting to the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

FIGURE 35. Input and output capacitance test circuit for table XX.

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